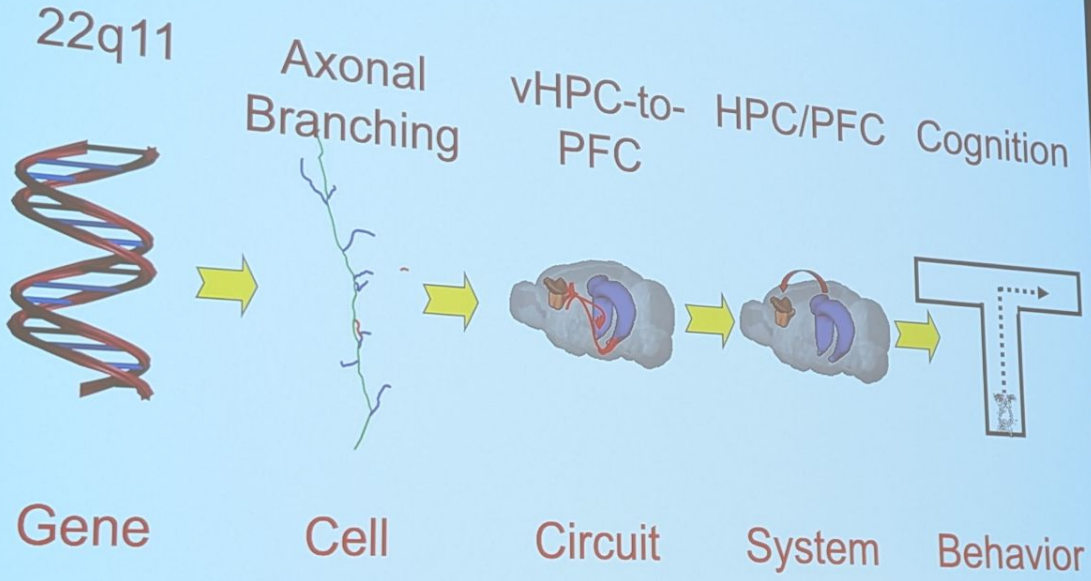
Data Repositories & Knowledgebases

across the BRAIN Initiative

MIT 2026-08-17

From Gene to Phenotype

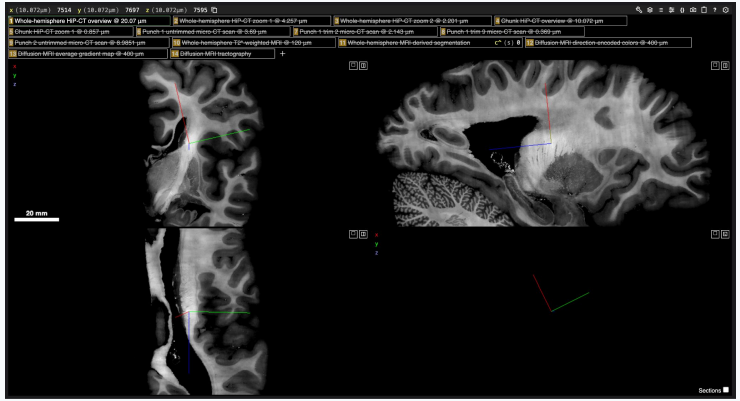


Sigurdsson et al., Nature, 2010; Spellman et al., Nature, 2015; Mukai et al., *Neuron*, 2015; Tamura et al., Neuron, 2016; Tamura et al., Nature Comm, 2017

BRAIN KNOWLEDGEBASE INNOVATION DOMAIN

The prompt

Create a flythrough of the cerebellum using this neuroglancer link:



https://gallery.lincbrain.org/mri-xray

NIH The BRAIN Initiative®

BICAN

BRAIN CONNECTS

BBQS

DANDI
https://dandiarchive.org

Reach Every Reader

ReadNet Project

SIMONS FOUNDATION

NRF National Research Foundation of Korea

MIT Generative AI Consortium
Tan-Yang ICON Center
Lann and Christopher Woehrle Psychiatric Research Fund

NIH National Institute of Biomedical Imaging and Bioengineering

ReproNim: A Center for Reproducible Neuroimaging Computation

NIH National Institutes of Health
Office of Strategic Coordination–The Common Fund

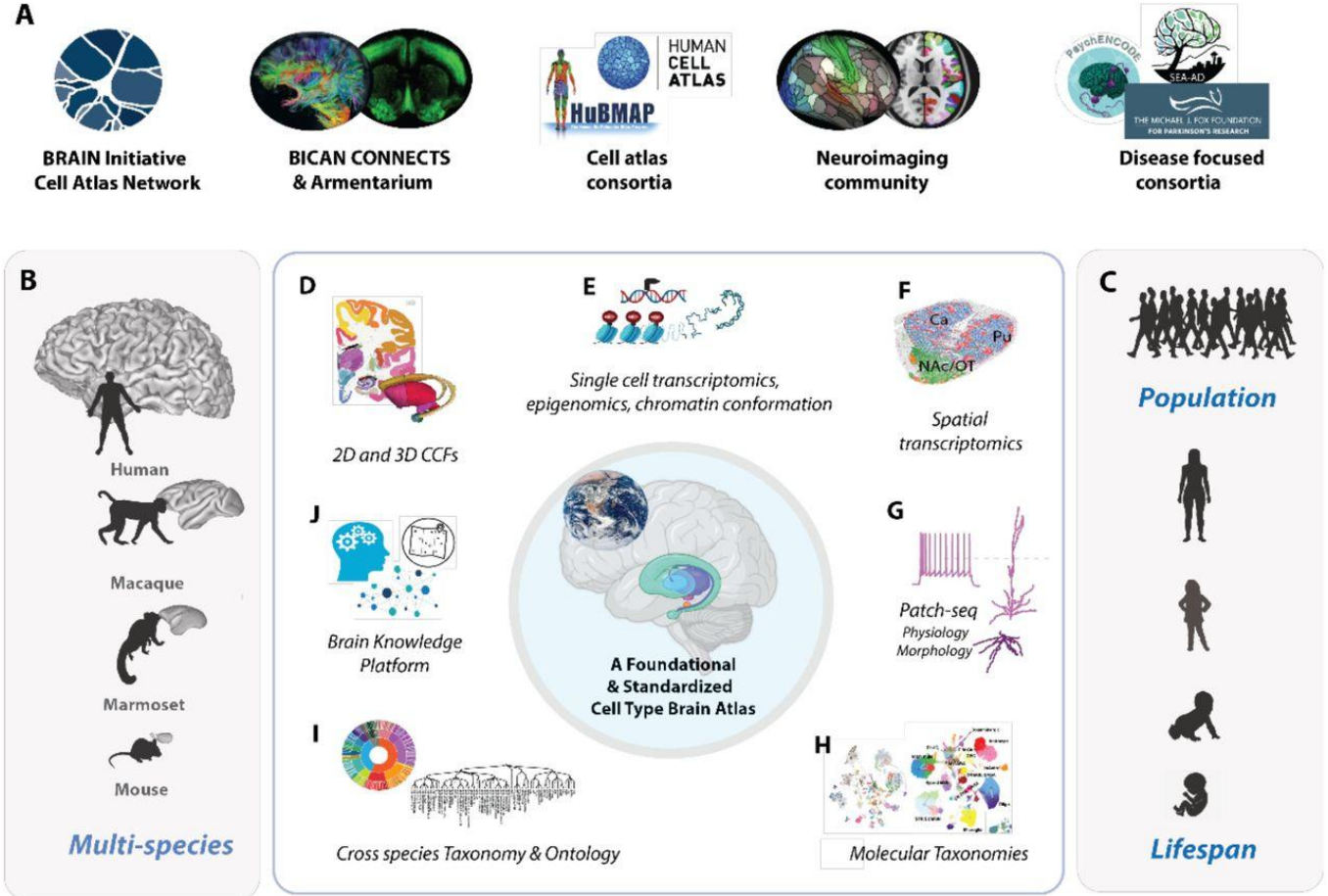
Bridge2AI Voice
Fueling Voice As Biomarker of Health

Child Mind Institute

Children's Hospital of Philadelphia®

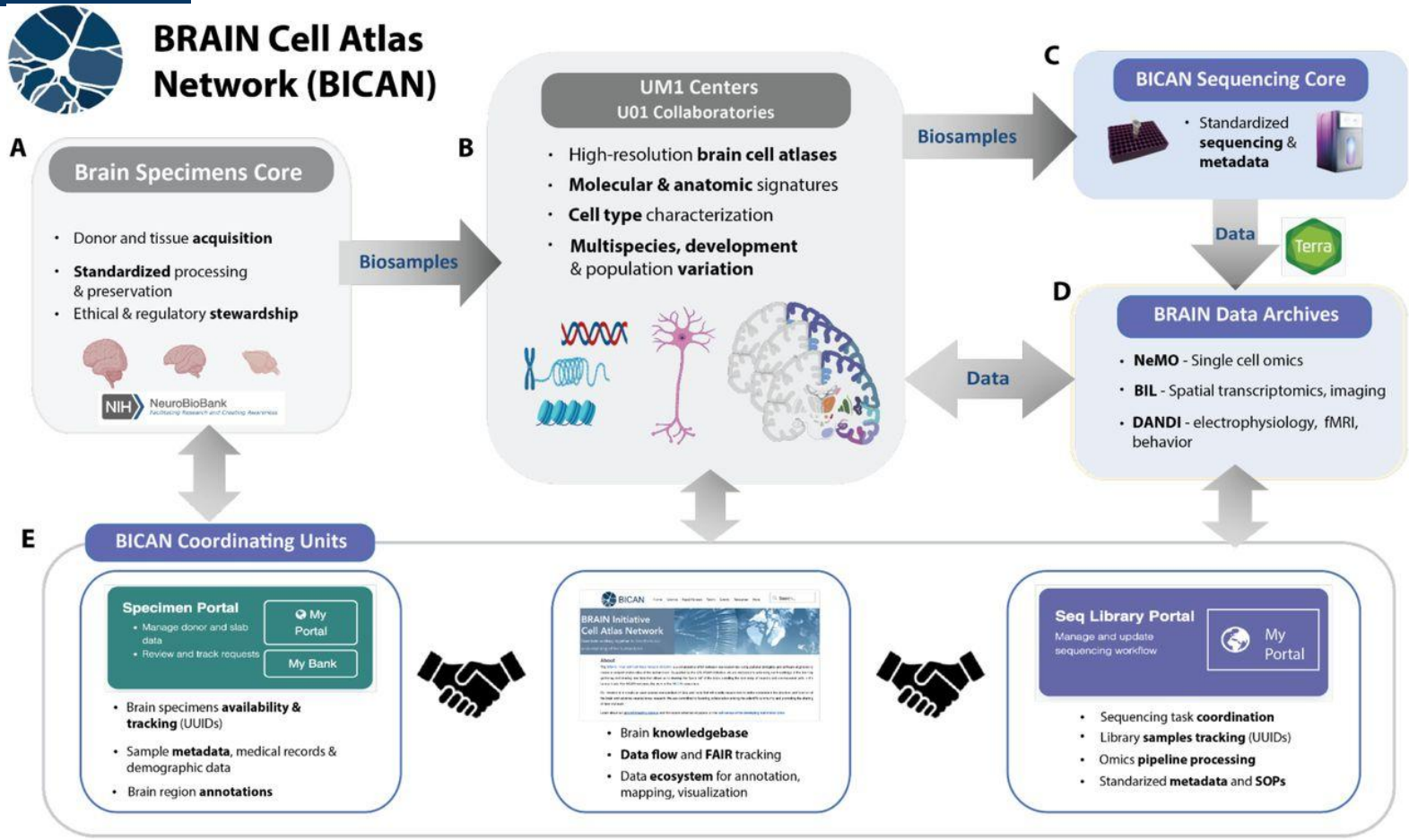
The BRAIN Initiative Projects

Brain Initiative Cell Atlas Network (BICAN)



https://www.biorxiv.org/content/10.64898/2026.04.14.717814v1.full

Brain Initiative Cell Atlas Network (BICAN)

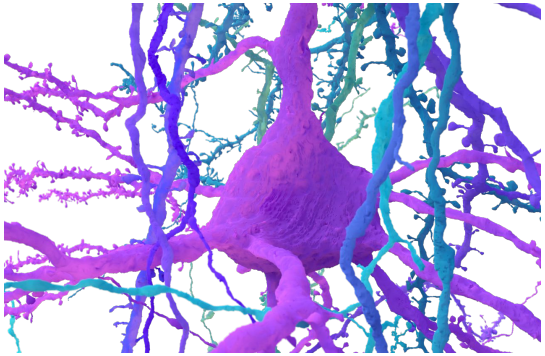


https://www.biorxiv.org/content/10.64898/2026.04.14.717814v1.full

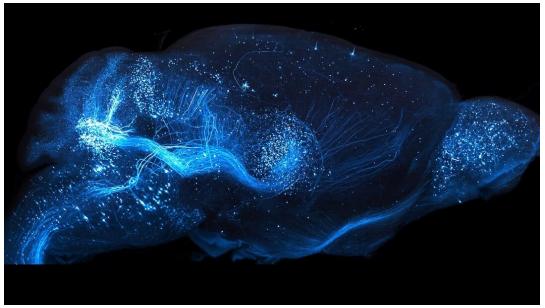
Circuits

BRAIN Initiative Connectivity Across Scales (CONNECTS)

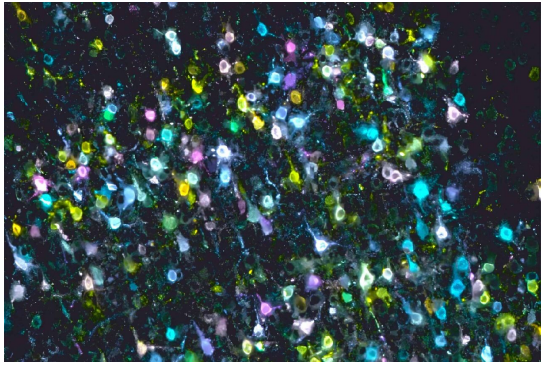
Electron microscopy projects



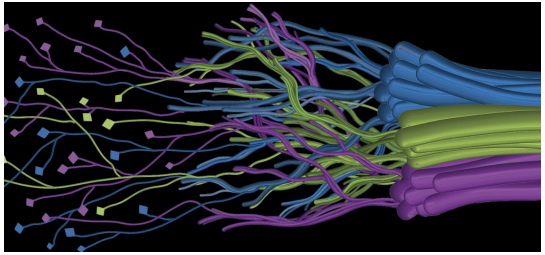
Lightsheet microscopy projects



Barcoded connectomics projects



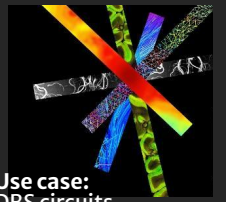
Fiber tractography projects

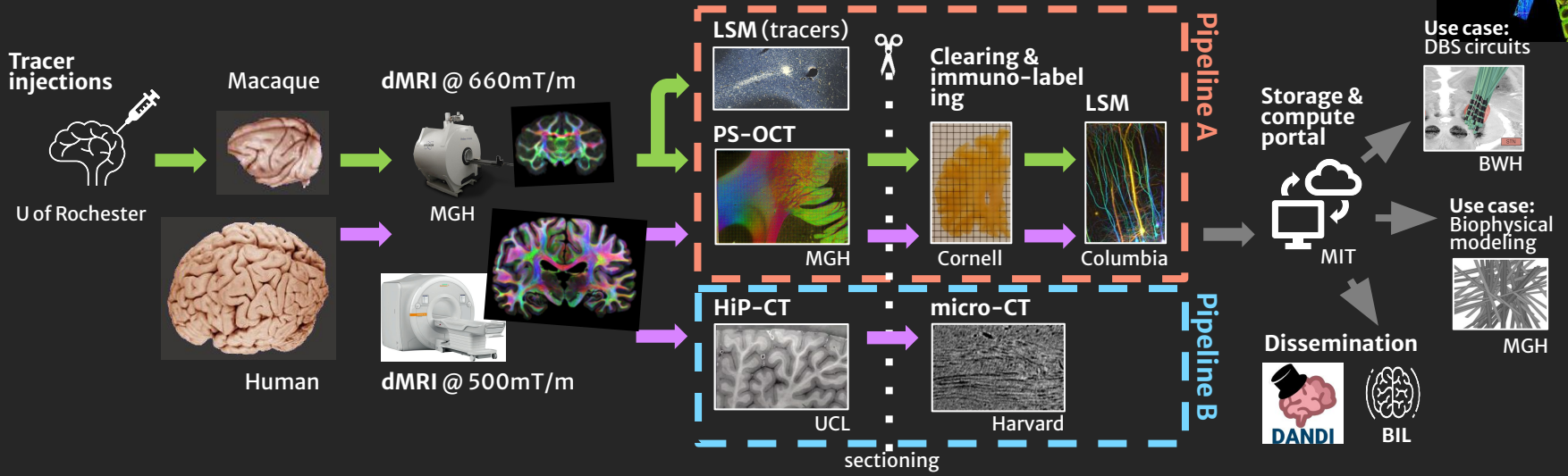


- UM1 Lichtman - EM
- UM1 da Costa - EM and ...
- U01 Kasthuri - EM
- U01 Rivlin - EM
- U01 Wanner - EM
- U01 Samuel - EM
- U01 Lee - EM
- U01 Fischl - LM
- UM1 Yendiki - LM
- U01 Sorensen - LM
- U01 Reid - LM
- UM1 Feng - Barcoded C...
- U01 Chen - Barcoded Co...
- U01 Macosko - Barcode...
- U01 Tolias - Barcoded C...
- U01 Duan - Barcoded Co...
- UM1 Ugurbil - Center for...
- U24 Pestilli - DCC
- U24 Zeng - DCC
- U01 Schaefer - X-ray

Circuits

Large-scale Imaging of Neural Circuits (LINC)











CONNECTS UM1
NS132358

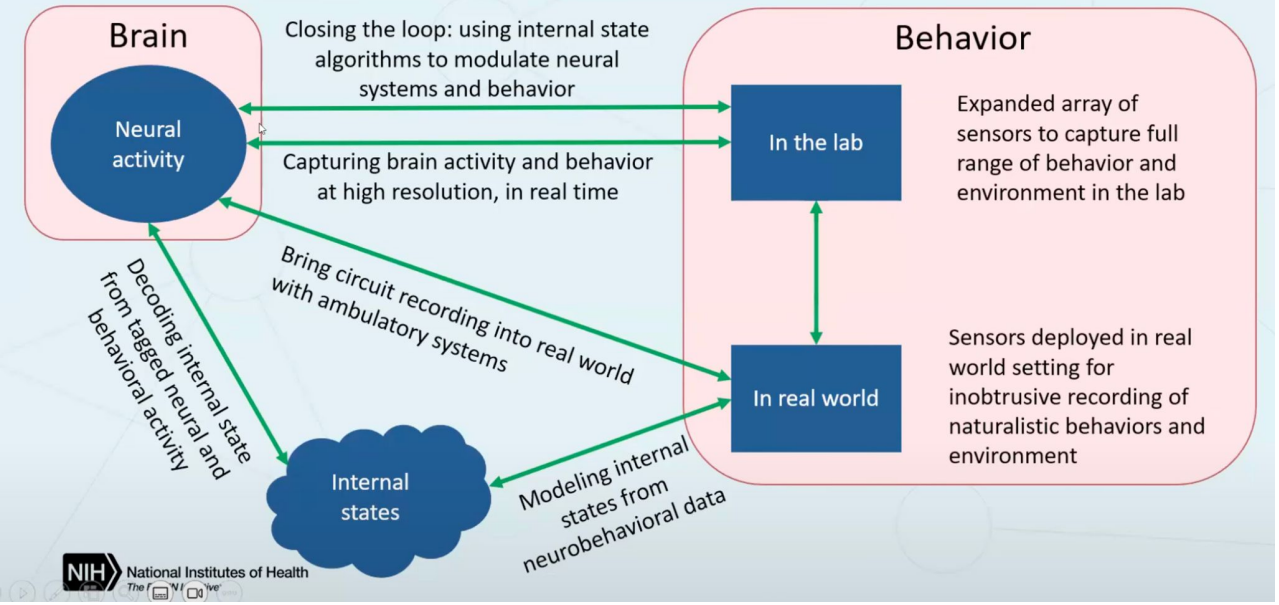
MPIs:
- Suzanne Haber (U of Rochester)
- Elizabeth Hillman (Columbia U)
- Anastasia Yendiki (MGH)

Site PIs:
- Mike Fox (BWH)
- Satra Ghosh (MIT)
- Jeff Lichtman (Harvard U)
- Claire Walsh / Peter Lee (UCL)
- Zhuhao Wu (Cornell Med)

Behavior

Brain Behavior Quantification and Synchronization (BBQS)

To gain a comprehensive mechanistic understanding of the neural basis of behavior (https://brain-bbqs.org)



- Developing tools to capture and quantify behavior with high temporal resolution and across multiple dimensions and across species
- Synchronizing with simultaneously recorded brain activity
- Building new conceptual and computational models that capture the complexity of behavior

- Establishing a cross-disciplinary consortium to develop, integrate, and disseminate tools, research designs, and ethical frameworks

Neuroethics and Responsible AI

- **Data collection methods**: engaging respectfully with donors and their communities, collecting heterogeneous cohorts of donors
- **Bias**: Questions, analysis, curation pipelines
- Ensuring **broad access** to software tools and archived data
- Good models are based on **good data**
  - Good data also bring context / provenance
- Neurotechnology / mental privacy / policy

SeeMe: A multimodal behavioral-electrophysiological tool for real-time detection of consciousness

Grant: 1R61MH138612

All PIs: Petar M Djuric, Charles B Mikell, Sima Mofakham

Capturing Autobiographical memory formation in People moving Through real-world spaces Using...

Grant: R61MH135109

All PIs: Cory Shields Inman

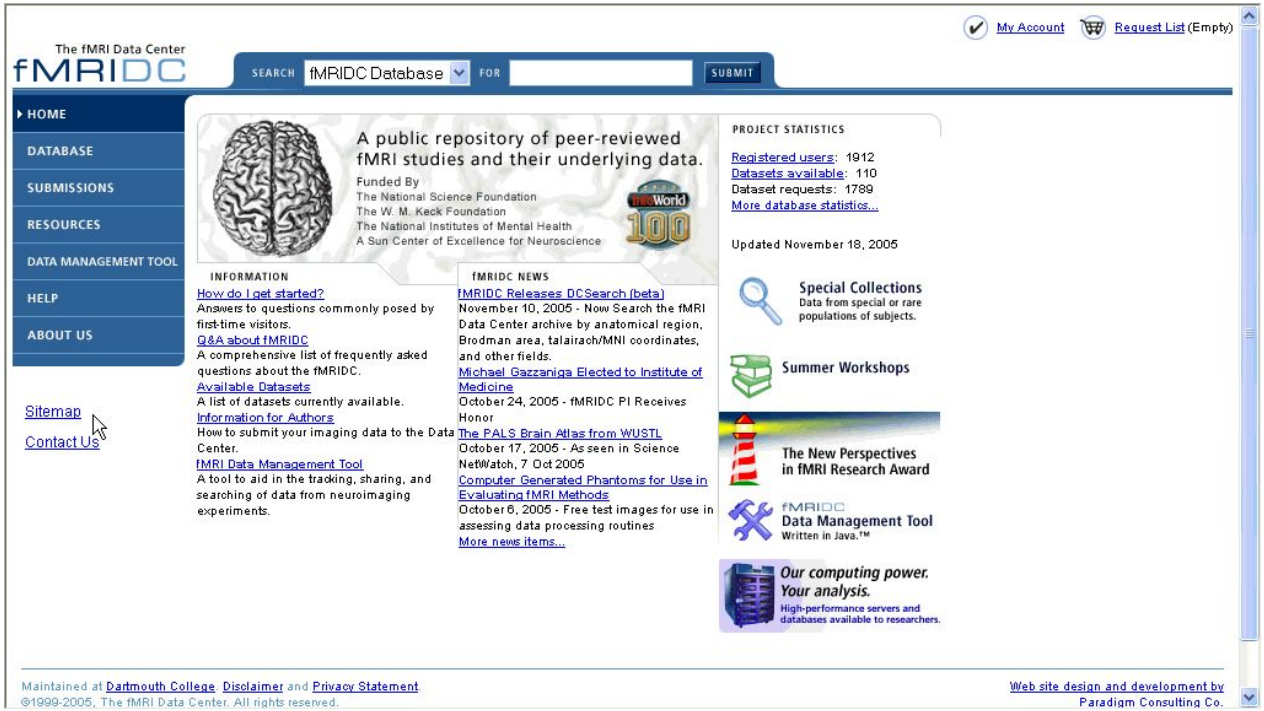
Novel multimodal neural, physiological, and behavioral sensing and machine learning for mental states

Grant: R61MH135407

All PIs: Maryam Shanechi

Data sharing and its future

A false start for fMRI data sharing



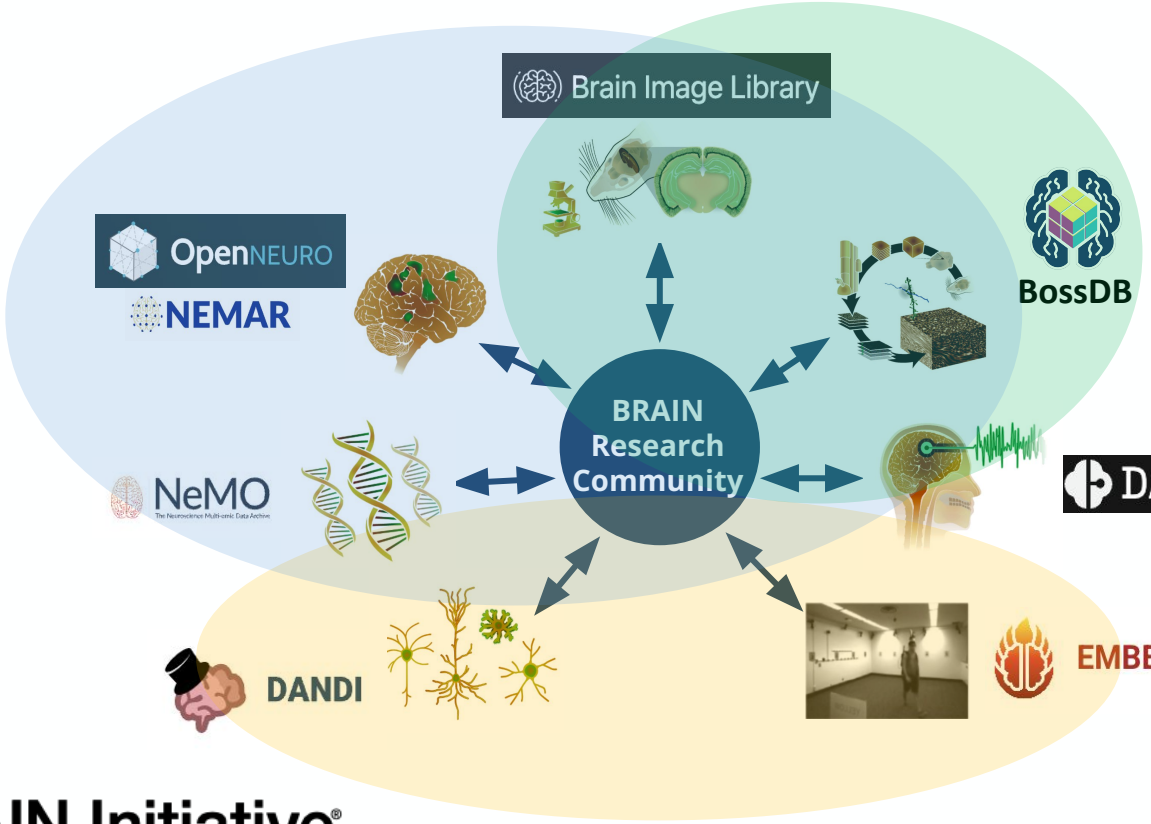
A false start for fMRI data sharing



This letter comes from a group of scientists who are publishing papers using fMRI to understand the links between brain and behavior. We are writing in reaction the recent announcement of the creation of the National fMRI Data Center (www.fmridc.org). In the letter announcing the creation of the center, it was also implied that leading journals in our field may require authors of all fMRI related papers accepted for publication to submit all experimental data pertaining to their paper to the Data Center. …
We are particularly concerned with any journal's decision to require all authors of all fMRI related papers accepted for publication to submit all experimental data pertaining to their paper to the Data Center.

BRAIN Knowledgebase ID

BRAIN Knowledgebase Domain focuses on Connecting Data to Accelerate Discovery



BICAN

BRAIN CONNECTS

BBQS
Brain-Behavior Quantification and Synchronization

The BRAIN Initiative®

Building the Foundation for Big Data Neuroscience

BRAIN's Innovative Path:

- **Start small, scale fast**: Risk-managed innovation
- **Community-driven approach**: Standards that researchers actually use
- **Domain expertise preserved**: Specialized archives, not generic storage

2014
- The U.S. BRAIN Initiative launched with a core objective to establish open and accessible data sharing platforms

2016
- The BRAIN informatics program pioneered an innovative approach to domain-specific repositories that preserve scientific expertise

2018
- Catalyzed community-driven data standards for neuroscience
- NWB 2.0 data standard released, BIDS support expanded

2020
- BRAIN's first-of-its-kind data policy for broad sharing takes effect (Jan 2020)
- Three years ahead of NIH-wide DMS policy (Jan 2023)

2023
- Cell-census network began transition to whole-brain atlas project
- Transformative BRAIN CONNECTS and BBQS programs launched

2024
- Exponential data growth across BRAIN archives achieved
- BRAIN-supported data standards widely adopted

Now
- Over 12 petabytes housed across BRAIN archives
- Working to enhance integration for maximizing data impact

Building on a Decade of Leadership in Brain-Scale Data

Established data standards ecosystem

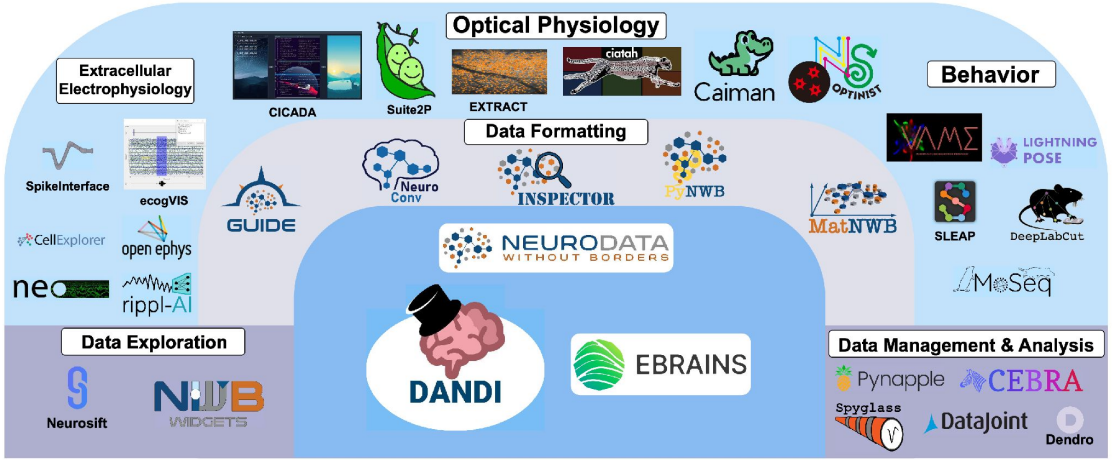
- Community-driven standards like NWB and BIDS now widely adopted across neuroscience
- Domain-specific standards development processes for cell-type taxonomies and physiology that can extend to connectivity mapping and behavioral analysis

Transformative project integration experience

- BICCN/BICAN mobilized 250+ researchers to create common reference frameworks and comprehensive brain cell atlases with coordinated data sharing
- CONNECTS and BBQS are building on and learning from BICAN's data management approaches

Active researcher communities

- Engaged ecosystem of data producers and consumers spanning all scales of brain investigation
- Established working groups for standards development and community-driven governance

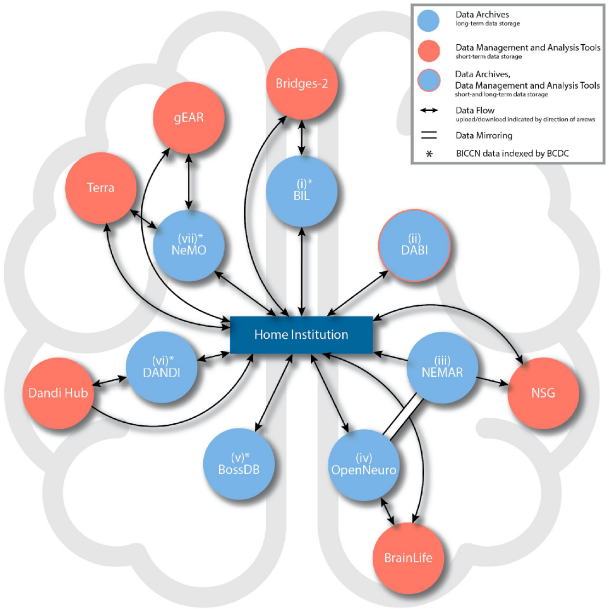


Gillon CJ, et al. ODIN: Open Data In Neurophysiology: Advancements, Solutions & Challenges. arXiv. 2024. http://arxiv.org/abs/2407.00976

"Current" BRAIN Data Landscape: Reviews → Tracking

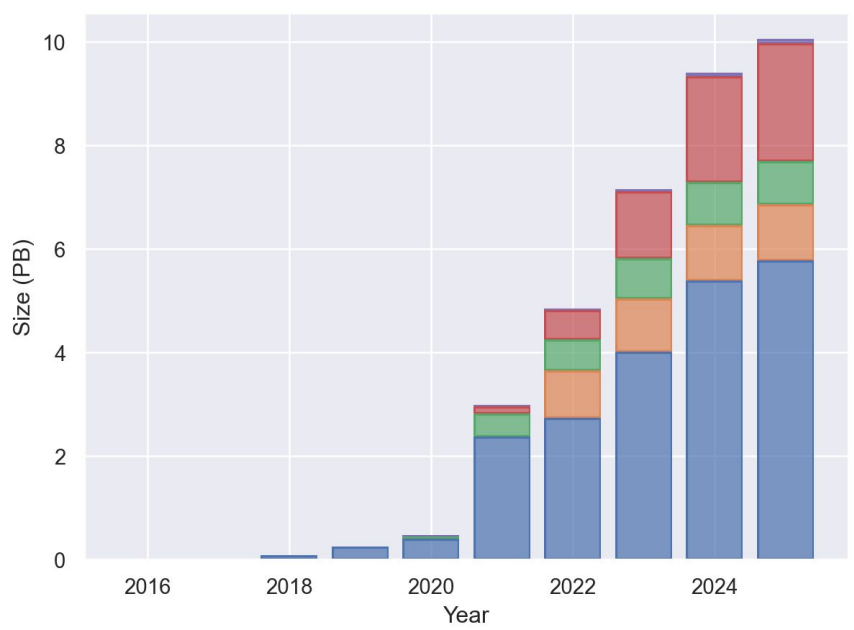
Table 1. BRAIN Initiative data archive ecosystem key statistics and related information.
Sources: For Host Institution, Initial Funding Date, Total NIH Support, Administering ICO, and Funding ICO, information was taken from the NIH RePORTER entry for the most recent award for each archive, as of June 6, 2024 (https://reporter.nih.gov/) Total NIH Support for each archive was taken from the History section of RePORTER. A separate line of NIH funding, totaling $1,707,134, with NIMH listed as the awardee, has supported the development of OpenNeuroPET. Other information in the table was derived from the International Neuroscience Coordinating Facility (INCF) Infrastructure Portfolio (https://www.incf.org/infrastructure-portfolio) and archive websites. Where an archive landing page or its data portal did not offer a public 'datasets' (or 'dandisets', in the case of DANDI) count, we used an available alternative such as the number of 'projects' for which data are available. *Data current as of June 6, 2024. ICO = NIH Institutes, Centers, and Offices.

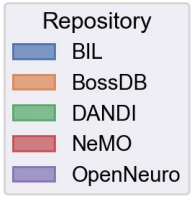
Archive	Host Institution	Initial Funding Date	Total NIH Support (USD)*	Administering ICO	Funding ICO	Data Types Hosted	Data Formats Supported	Publicly Available Datasets*
Brain Image Library (BIL)	Carnegie-Mellon University	2017	$6,767,625	NIMH	NIMH	Confocal microscopy	DICOM or NIfTI; for derived data, any format as long as primary data are in the required format	8418
Brain Observatory Storage Service and Database (BossDB)	Johns Hopkins University	2018	$5,133,219	NIMH	NIMH	Electron microscopy; x-ray microtomography	PNG; APNG; JPG; BMP; GIF; PSD	50
Data Archive for the BRAIN Initiative (DABI)	University of Southern California	2018	$6,160,745	NIMH	NINDS	Focus on invasive neurophysiology; houses all imaging and brain signal data	EDF, Biosemi (.bdf), BrainVision (.eeg,.vhdr,.vmrk), EEGLAB (.set,.fdt), Blackrock NeuroPort (.nev,.nsX), Intan (.rhd,.rhs), MATLAB files (.mat,.m); iEEG-BIDS, NWB, or DABI	110
Distributed Archives for Neurophysiology Data Integration (DANDI)	Massachusetts Institute of Technology	2019	$8,309,010	NIMH	NIMH	Cellular neurophysiology, neuroimaging, and microscopy	BIDS; NWB	640
NeuroElectroMagnetic data Archive and tools Resource (NEMAR)	University of California, San Diego	2019	$4,464,874	NIMH	NIMH	EEG, MEG	BIDS	297
Neuroscience Multi-Omic Data Archive (NeMO)	University of Maryland, Baltimore	2017	$9,347,683	NIMH	NIMH	Multi-omics	FASTQ; TSV; BAM; BIGWIG; MEX; LIST; BED; QBED; CSV; H5; BIGBED; LOOM; MTX; ASPX	49
OpenNeuro	Stanford University	2018	$7,254,848	NIMH	NIMH	MRI, PET, MEG, EEG, iEEG	BIDS	1076

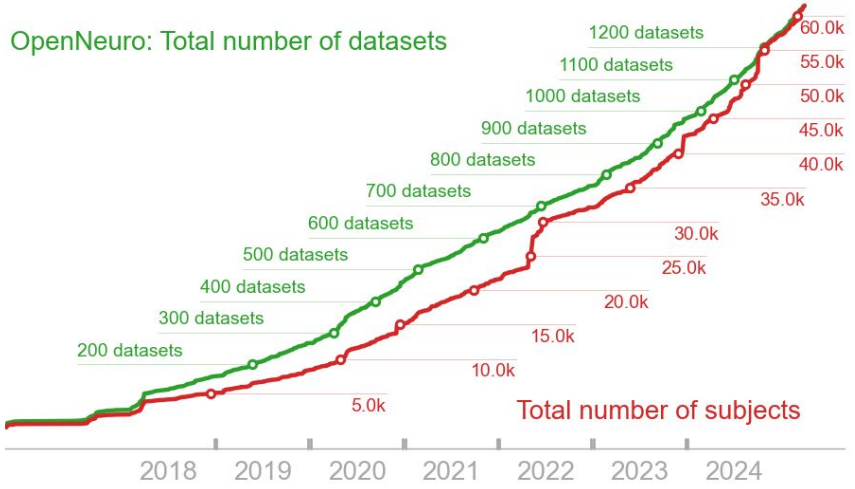


Iyer S, et al. The BRAIN Initiative data-sharing ecosystem: Characteristics, challenges, benefits, and opportunities. Elife. 2024;13: e94000. doi:10.7554/elife.94000

Open Neuroscience Data
BRAIN Initiative Repositories







NeMO
The Neuroscience Multi-omic Archive

- *Transcriptomics, Genomics*
- *Epigenomics*
- *Spatial Transcriptomics*
- *Patch-seq*

nemoarchive.org

BIL
BRAIN Image Library

- *Morphology / Connectivity*
- *Microscopy, Imaging*
- *Spatial Transcriptomics*
- *Patch-seq*

www.brainimagelibrary.org

DANDI
Distributed Archives for Neurophysiology Data Integration

- *Electrophys & Optophys*
- *Behavior, Stimulation*
- *Microscopy & MRI*
- *Patch-seq*

dandiarchive.org

BossDB
Brain Observatory Storage Service & Database

- *X-ray Tomography*
- *Electron Microscopy*
- *Connectivity*

bossdb.org

OpenNeuro
OpenNeuro

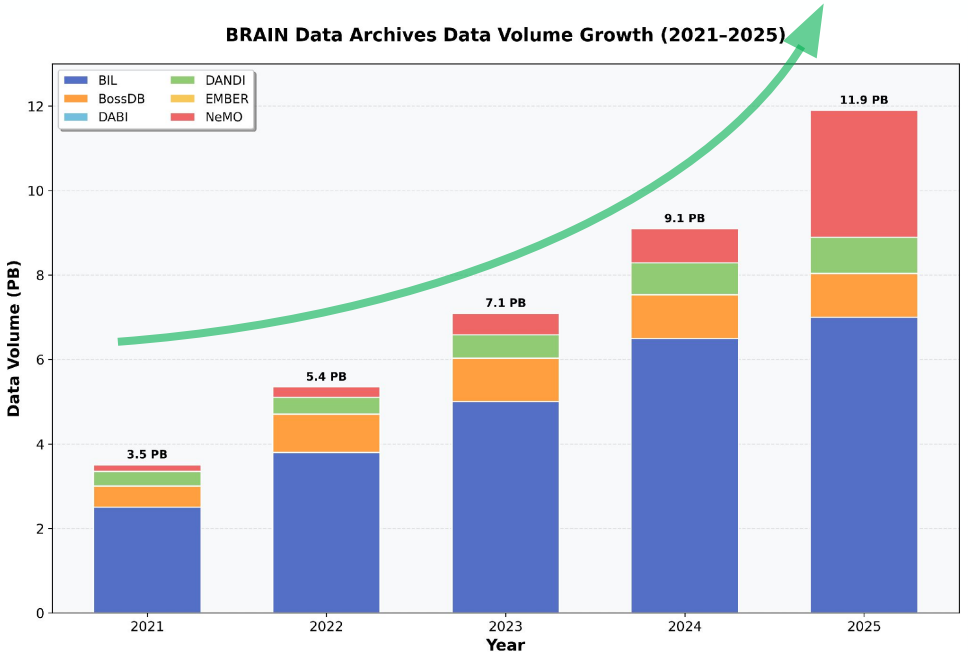
- *MRI*
- *EEG/MEG/iEEG*
- *PET*

openneuro.org

Exponential Growth: Success to Sustainability

Success Creates New Challenges:

- **Computational burden**: Analyzing petabyte-scale datasets
- **Sustainability pressures**: Storage costs growing exponentially
- **Cross-modal integration**: Connecting insights across archives
- **Discovery barriers**: Finding relevant data across scales, projects, and platforms
- **Reproducibility demands**: Maintaining scientific quality at scale



Data Management and Sharing (DMS) Policy: Updates

NIH DMS and BRAIN DMS Policy

- NOT-MH-19-010: Notice of Data Sharing Policy for the BRAIN Initiative
  - Effectively deprecated by NOT-OD-26-046 as of May 25, 2026
- NOT-OD-21-013: Final NIH Policy for Data Management and Sharing
- NOT-OD-26-046: Updated Elements of an NIH Data Management and Sharing Plan
  - Updated Elements of an NIH Data Management and Sharing Plan
  - New DMS Plan template: https://grants.nih.gov/sites/default/files/dms-blank-format-page-2026.docx
- NIH DMS | Frequently Asked Questions (FAQs) | Grants & Funding

BRAIN Website: Previous vs. New (Work-In-Progress)

- Data Science and Informatics | BRAIN Initiative (Previous)
- Informatics Program | BRAIN Initiative (Previous)
- Data and Knowledge Resources | National Institutes of Health (NIH) (New IC/O Webpage; WIP)

List of Data Archives

BRAIN Data Archives

The Initiative has established a network of domain-specific data archives serving particular data types. Investigators should submit data to the archive that matches their data type and modality:

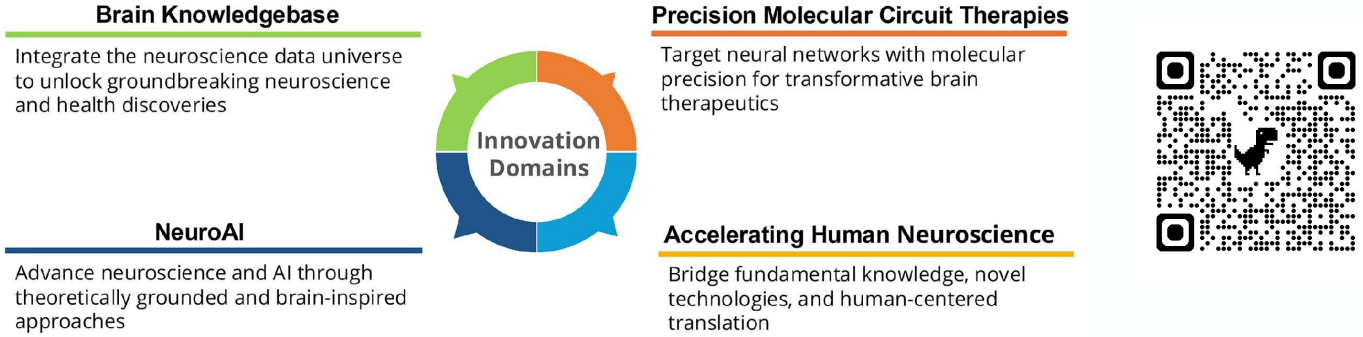
Archive	Data Types	Link
DANDI	Cellular neurophysiology, electrophysiology, optophysiology, behavioral time-series	dandiarchive.org
NeMO	Multi-omic data (transcriptomics, epigenomics, spatial omics)	nemoarchive.org
DABI	Human participant intracranial electrophysiology	dabi.loni.usc.edu
OpenNeuro	Human neuroimaging (MRI, fMRI)	openneuro.org
OpenNeuro PET	Positron emission tomography	openneuropet.github.io
NEMAR	Human EEG, MEG, iEEG (shared through the OpenNeuro portal)	nemar.org
BIL	Large-scale microscopy brain imaging (confocal, light sheet)	brainimagelibrary.org
BossDB	Electron microscopy and X-ray microtomography	bossdb.org
EMBER	Multimodal behavioral and neural data	brain-bbqs.org

Investigators working within BRAIN Initiative consortia (BICAN, CONNECTS, BBQS) should coordinate data submissions through their consortium's data coordinating center.

BRAIN Data and Knowledge Resources: Data Archives Table
https://www.nih.gov/brain-initiative/research/data-knowledge-resources

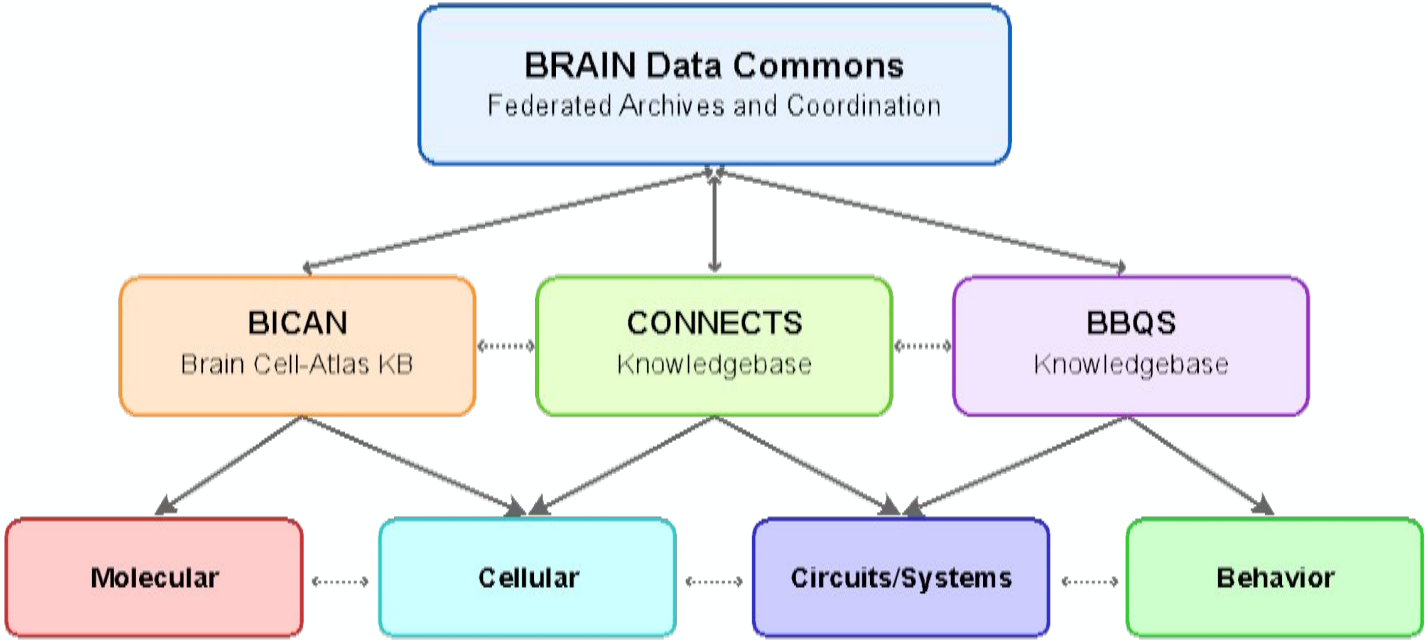
BRAIN Initiative Research Roadmap: Inventing the Future

The NIH BRAIN Research Roadmap identifies four Innovation Domains designed to build on the BRAIN Initiative's first decade of progress and accelerate the next generation of neuroscience discovery.



- NeuroView, "Inventing the future: A Neuroscience research roadmap", *John Ngai, Neuron, May 2026*
- NINDS Director's message "Inventing the Future of Neuroscience: A Shared Roadmap for the Next Decade"

BRAIN Knowledgebase Domain focuses on Connecting Data to Accelerate Discovery



Guided by core principles: Modularity, Progressive Data Maturity, Balanced Decentralization, and Collaborative Integration.

The repositories

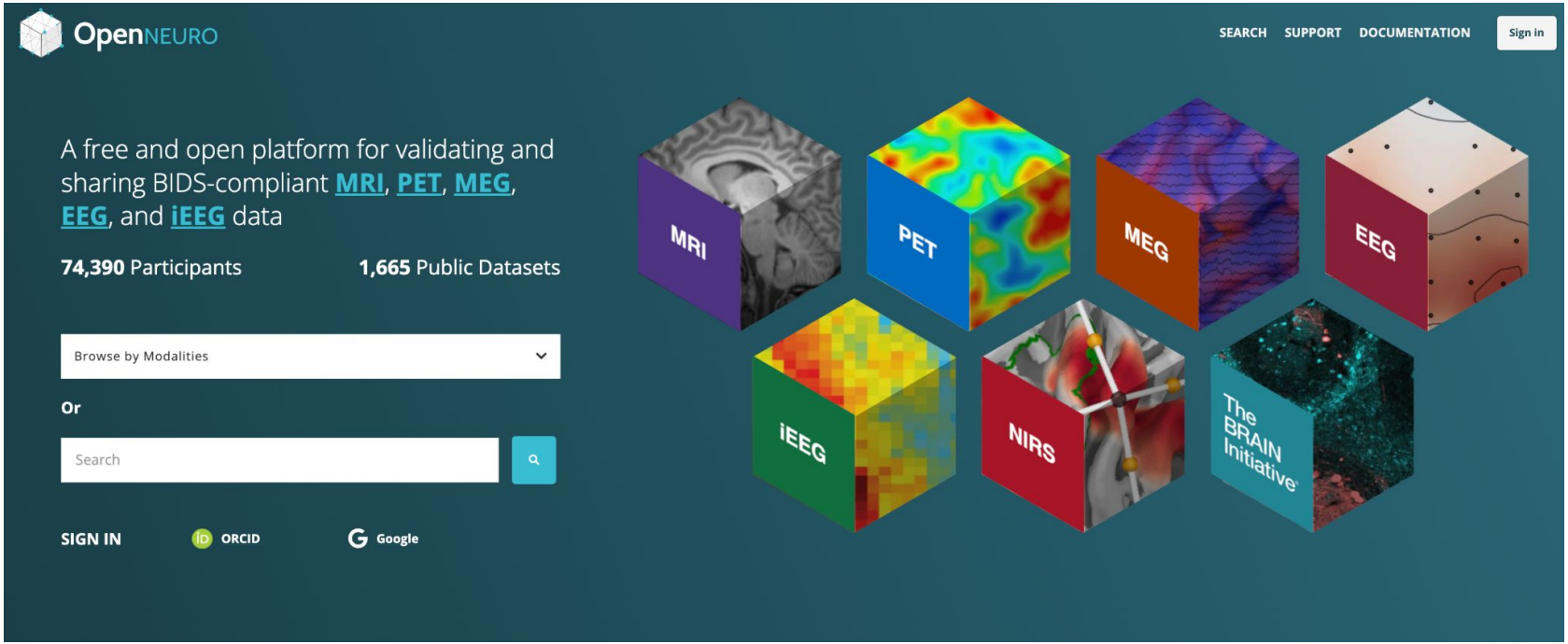
DATA SHARING FOR OPEN SCIENCE



DATA REUSE FOR SCIENTIFIC DISCOVERY



OpenNeuro: Sharing raw and processed neuroimaging data



Markiewicz et al, 2021, *eLife*

An open science ecosystem for neuroimaging

OpenNEURO BETA
A free and open platform for analyzing and sharing neuroimaging data

NEUROVAULT
A public repository of unthresholded statistical maps, parcellations, and atlases of the human brain

neurosynth.org

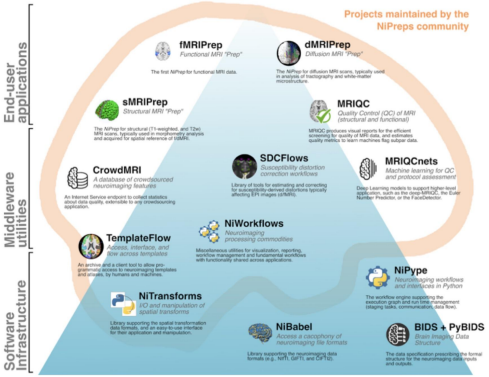
Neurosynth is a platform for large-scale, automated synthesis of functional magnetic resonance imaging (fMRI) data.

It takes thousands of published articles reporting the results of fMRI studies, chews on them for a bit, and then spits out images that look like this:



An automated meta-analysis of 885 studies of language

Open data



Open source software

BIDS
BRAIN IMAGING DATA STRUCTURE

Open standards

The Legacy NEMAR website is available at https://ww1.nemar.org until September 30th, 2026.

NEMAR

About Discover Citation Dashboard Documentation Support Sign in

OPEN NEUROSCIENCE DATA, READY TO USE

Search for data

Enter keyword or phrase

Search

Search, Visualize, Analyze, Download

NEMAR is an open-access *data, tools, and compute* resource for finding, accessing, assessing, and processing human EEG, MEG, iEEG, and EMG data — contributed by researchers, mirrored from OpenNeuro, and BIDS-formatted with data-quality and HED event annotations.

Browse datasets

Contribute your dataset →

754
Datasets hosted

39K
Participants

57.2 TB
Hosted data

BRAIN PET ANALYSIS PIPELINES

Automate Data Processing

LEARN MORE GET PETPREP

ABOUT US

We are neuroscientists just like you

BRAIN SCIENCE

OPEN SCIENCE

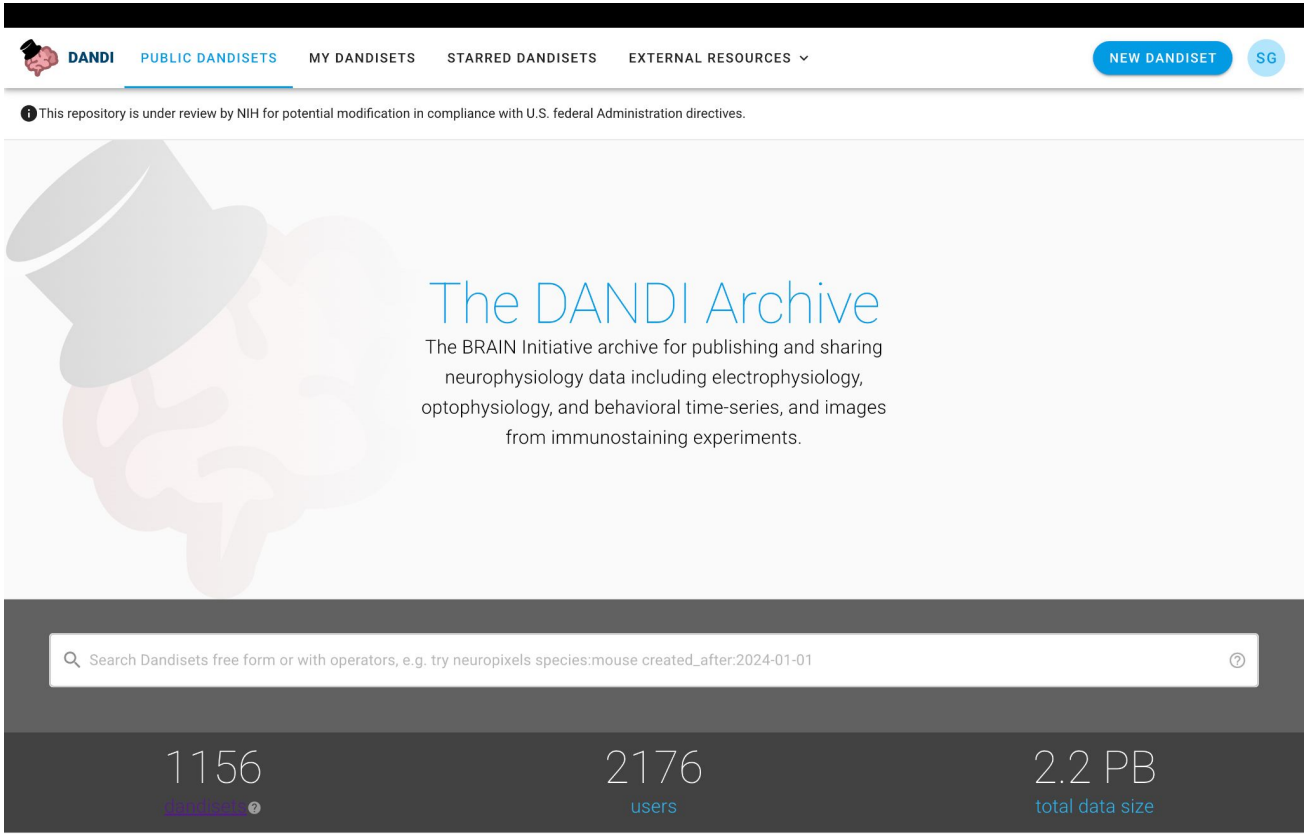
REPRODUCIBILITY

WEB-CENTRIC

Research findings can be difficult to replicate, partly because of inconsistencies in data acquisition, preprocessing, and analysis. Our research seeks to understand the source of these discrepancies for PET imaging, focusing primarily on standardization: how are neuroimaging data acquired, documented, analysed, and archived. Such work helps establish data sharing pipelines, which in turn enables replication and increases sample sizes.

The primary objective of OpenNeuro PET is data sharing, either Openly with OpenNeuro (CC0 licence) or Pulicly with Public-nEUro (Data access

A Collaboration Ecosystem for Neurophysiology Projects



DANDI
dandiarchive.org

NIH The BRAIN Initiative®

NIH National Institute of Mental Health

1R24MH117295

AWS Public dataset program

BICAN

BRAIN Connects

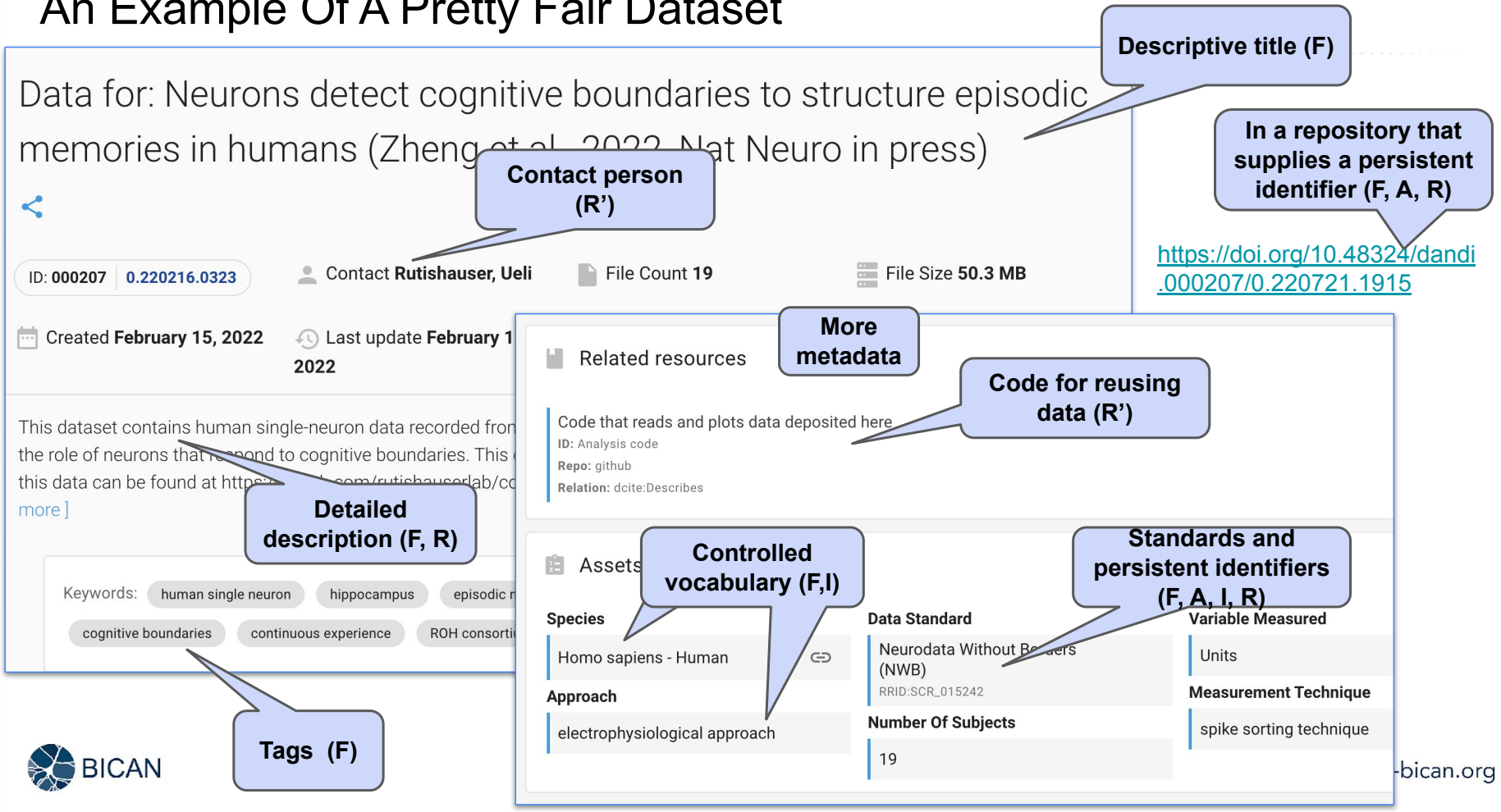
CATALYST NEURO

DARTMOUTH

Kitware

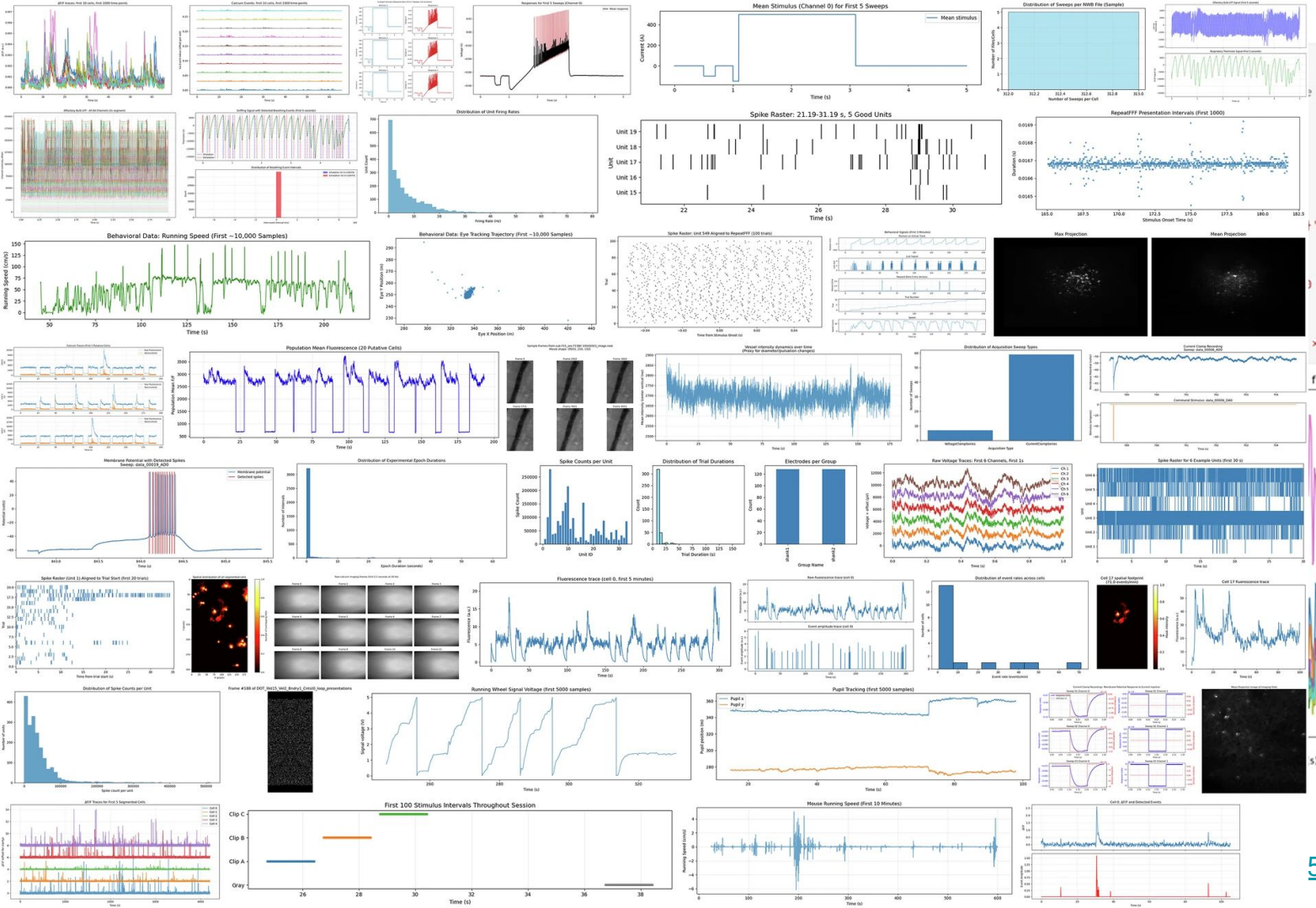
MIT

An Example Of A Pretty Fair Dataset



BICAN

-bican.org



points, first 10 cells

') time points')

x cells)")

first 1000 time points

s)

597-025-06285-x)

NeMO - The Neuroscience Multi-omic Data Archive

NeMO is a data repository focused on the storage and dissemination of -omics data generated from the BRAIN Initiative, SCORCH consortium and other brain research projects. The archive hosts more than 15 million files totaling ~3 petabytes of sequence data with collectively over 7 petabytes of downloads from our various download mechanisms.

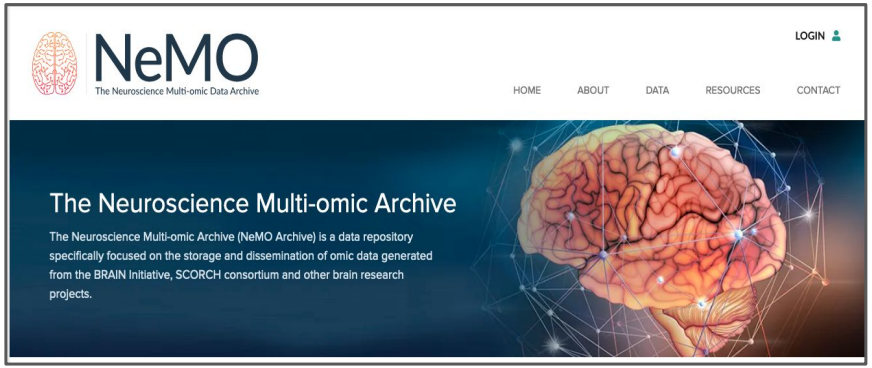
NeMO data includes:

Modalities	Transcriptome (including Spatial Transcriptome), Genome, Epigenome, Multimodal
Techniques	> 80 types
Organisms	> 25 species
Data Types	Sequence reads, demultiplexed fastqs, QC reads, alignment, counts, run metrics, analysis metrics etc.
File formats	> 25 formats

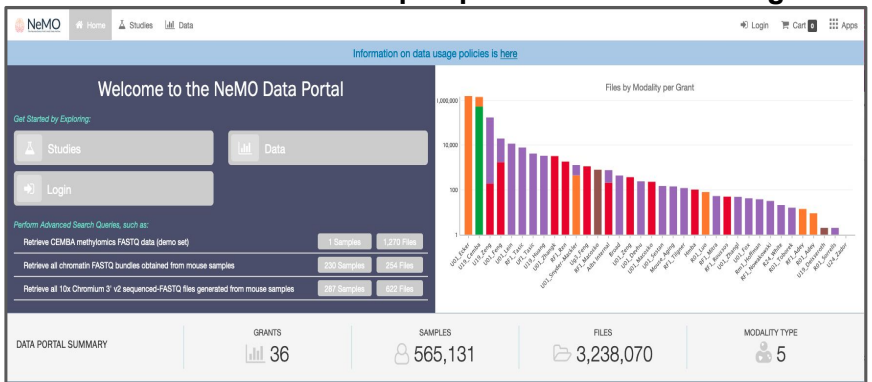
Data Submission: File manifest via Dashboard/API and Data files via Aspera
Accessing Data: Landing pages, HTTPS, GCP buckets (public and restricted data), and Portal Client.
Important Links:
Registering with NeMO
Data submission instructions
Accessing public data
Accessing restricted data

Contact NeMO: nemo@som.umaryland.edu

NeMo Dashboard: https://nemoarchive.org/



NeMO Portal: https://portal.nemoarchive.org/



Brain Image Library (BIL)

What is BIL?

- Centralized large-scale repository for brain microscopy data
- Integrated HPC resources for in-place data analysis & exploration.
- >11,000 datasets (GB-TB scale) 133M+ public files, >5PB total.

Data Types Supported

- Wide range of species
- Anatomy/Morphology: tracing, virus labeling, MORF, etc.
- Cell morphology: *neuron reconstructions*
- Imaging: *confocal microscopy, lightsheet, STPT, fMOST*
- **Multimodal:*Patch-seq***
- **Image Based Spatial Transcriptomics: *MERFISH, Bar-seq, FISH, etc***

Submission Process

- No data size limit
- Formats: Open format
  - Preferred: TIFF, OME-TIFF, OME-Zarr
  - These formats are supported by neuroglancer visualization
- Not limited to BICAN

Networking and Data Transfer Support

Data Transfer: SFTP/SCP, Globus, Rsync
Support including network analysis to find transfer bottlenecks.

Integrated Analysis Ecosystem for Pre/Post Data Publication Analysis

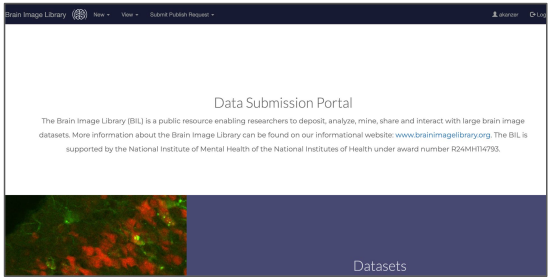
- Direct data access on HPC.
- Utilization of Open OnDemand for computation in JupyterLab, RStudio, VSCode, and remote desktop.
- Exploration, visualization, and seamless data access without download after publication.
- No cost to use any HPC resources

Get Started

Submission Instructions
Create an Account
Publications (Group DOI)

Support

Email: bil-support@psc.edu



Submission Portal:
https://submit.brainimagelibrary.org

Start the data submission process well in advance of any deadlines that you may have for making the data public

Filter by

Search in 172 projects

Showing 50/172 (Total: 172)

Sort by: Default

- Species (25)
- Modality (13)
- Technique (67)
- Primary Author (137)
- Specimen Type (19)
- Program (5)
- Sub Program (56)

Summary	Species	Modality	Technique	Primary Author
Seattle Alzheimer's Disease Brain Cell Atlas (SEA-AD) - 10x single nucleus RNAseq The Seattle Alzheimer's Disease Brain Cell Atlas (SEA-AD) is a consortium focused on gaining a deep molecular and cellular understanding of the early pathogenesis of Alzheimer's... 7 ABC Atlas Web Resource Protocols MapMyCells	human	transcriptomics	10x Chromium 3' v3 sequencing, 10x Chromium 3' v3.1 sequencing	Ed Lein, Allen Institute for Brain Science
Highly specific, renewable, and cost-effective antibody toolbox for 3D proteomic phenotyping of the brain System-wide analysis of cell types in the brain is essential for understanding how complex cellular interactions give rise to various brain functions. Rapidly evolving tissue transformatio... 1 Web Resource	human, mouse	histology, imaging	confocal microscopy	Kwanghun Chung
Toward a Human Adult Brain Cell Atlas with Single-Cell Technologies Towards producing a complete cell atlas of the human brain, including a full molecular catalog of all its cell types and their spatial distribution, we propose interrogating the human brai... 2 Protocols	human	epigenomics, multimodal, transcriptomics	10X Genomics Multiome, ATAC-seq, RNAseq	Bing Ren
Allen Mouse Brain Connectivity Atlas - Transgenic Characterization Transgenic mice are essential tools for studying cell types,	mouse	histology, imaging	in situ hybridization	Allen Institute for Brain Science

This repository is under review for potential modification in compliance with Administration directives.

BossDB

Brain Observatory Storage Service & Database

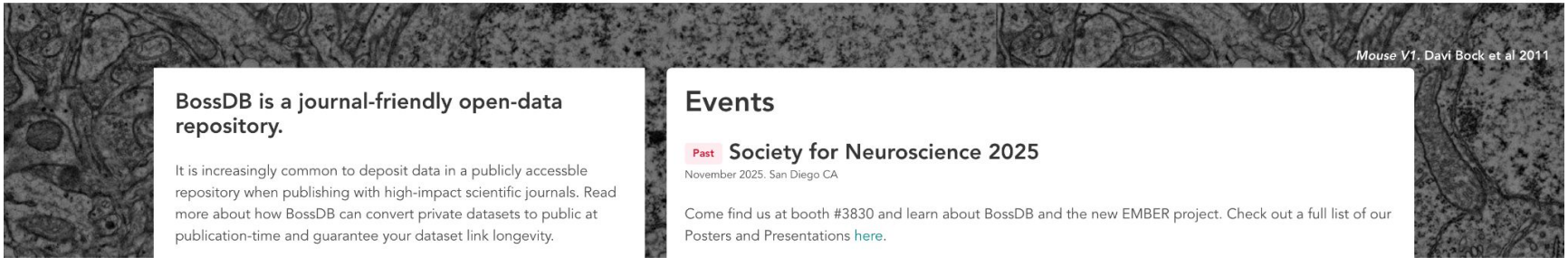
RRID:SCR_017273

BossDB is a volumetric database for 3D and 4D neuroscience data, focused on Electron Microscopy and X-ray Micro/Nano Tomography.

Store petabytes of high-dimensional data faster and bigger. BossDB currently stores thousands of terabytes of data from all over the world.

Browse Public Data | Submit Your Data | Get Started with Python

Current Database Size	1.4 PB
Projects	63



Mouse V1. Davi Bock et al 2011

BossDB is a journal-friendly open-data repository.

It is increasingly common to deposit data in a publicly accessble repository when publishing with high-impact scientific journals. Read more about how BossDB can convert private datasets to public at publication-time and guarantee your dataset link longevity.

Events

Past Society for Neuroscience 2025

November 2025. San Diego CA

Come find us at booth #3830 and learn about BossDB and the new EMBER project. Check out a full list of our Posters and Presentations here.

This repository is under review for potential modification in compliance with Administration directives.



EMBER Archive

Ecosystem for Multi-modal Brain-behavior Experimentation and Research

EMBER is the BRAIN Initiative data archive for multi-modal neurophysiological and behavioral data, supporting the Brain Behavior Quantification and Synchronization (BBQS) Program.

LEARN MORE

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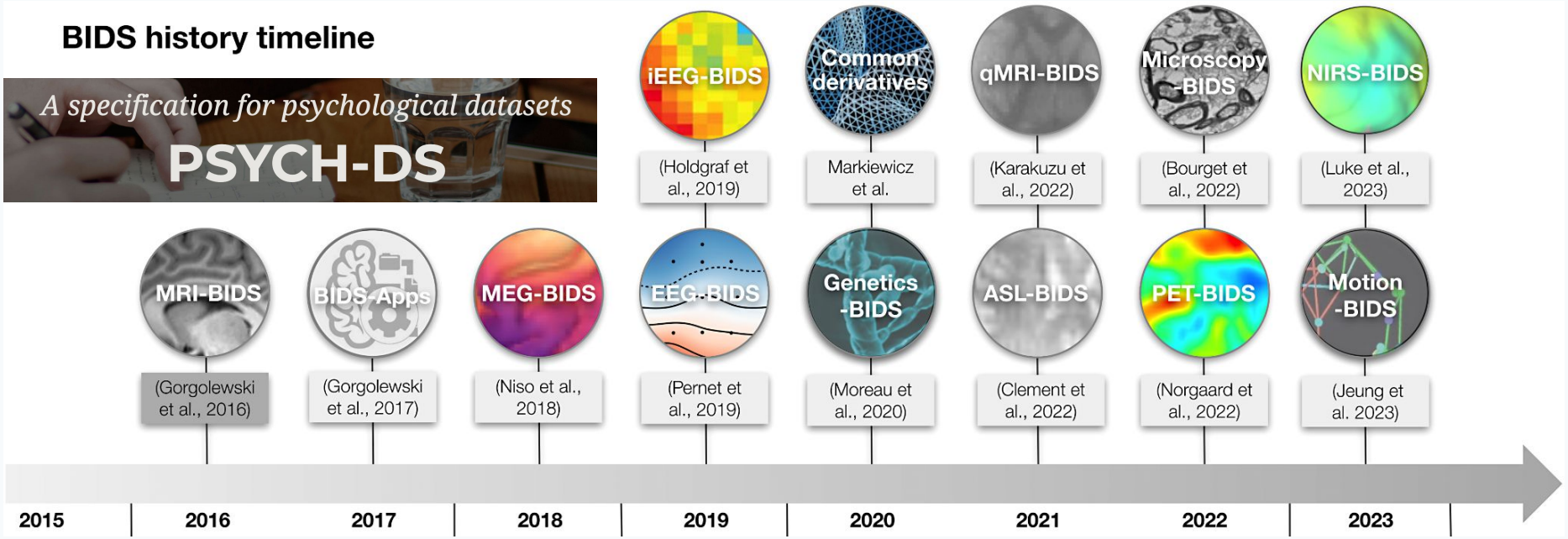
Standards, Pipelines, and Computation

NEURODATA WITHOUT BORDERS

Rübel et al. ELife (2022) https://doi.org/10.7554/eLife.78362

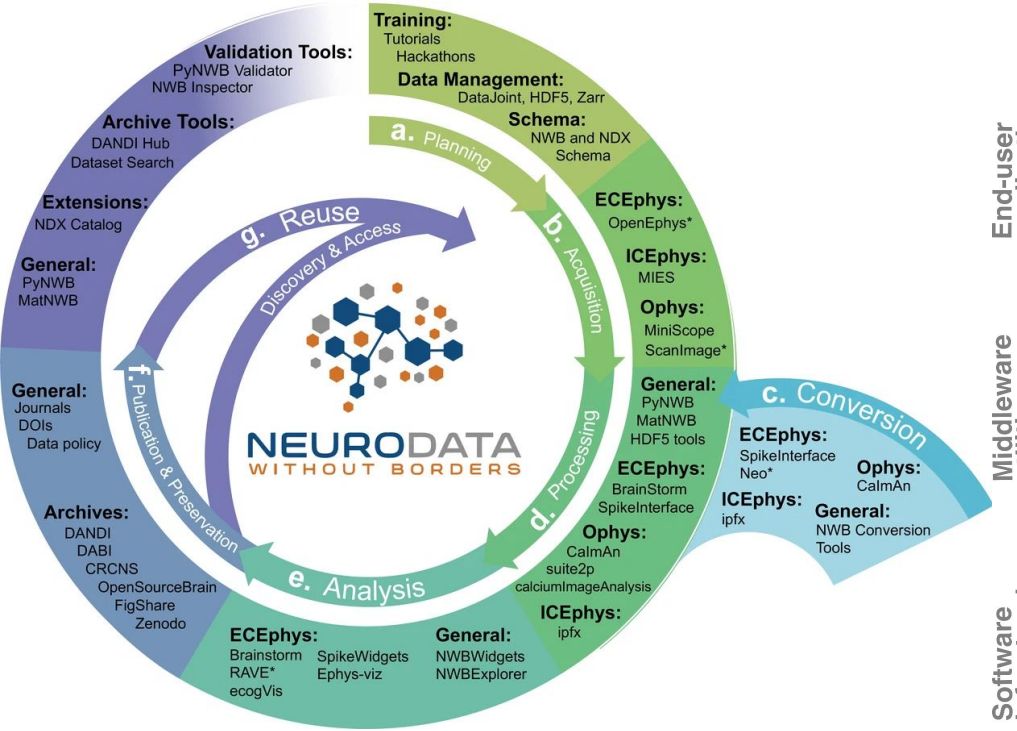
OME+Zarr

Moore at al.Histochem Cell Biol (2023)
https://doi.org/10.1007/s00418-023-02209-1



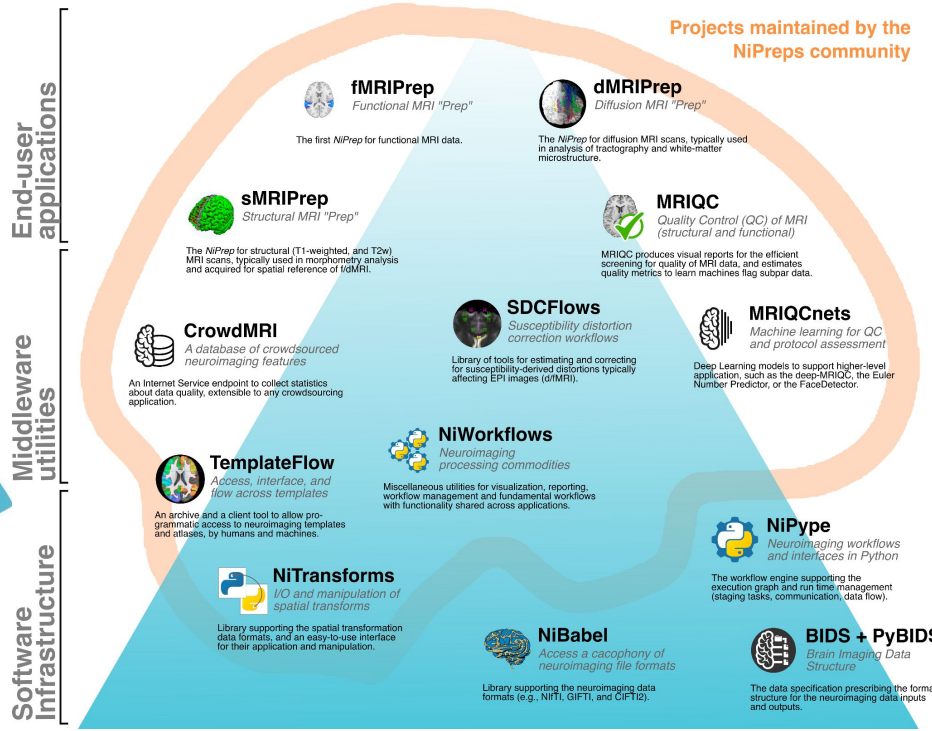
Poldrack et al. Imaging Neuroscience (2024)
https://doi.org/10.1162/imag_a_00103

Neurophysiology



Keles et al. (2024) Data for: Multimodal single-neuron, intracranial EEG, and fMRI brain responses during movie watching in human patients (Version 0.240227.2023) [Data set]. DANDI archive. https://dandiarchive.org/dandiset/000623/0.240227.2023

Neuroimaging



Keles et al. (2024). fMRI data for: Multimodal single-neuron, intracranial EEG, and fMRI brain responses during movie watching in human patients. OpenNeuro. [Dataset] doi: doi:10.18112/openneuro.ds004798.v1.0.5

Welcome to DANDI Compute: AIND Ephys

An experiment in reproducible electrophysiology processing on the DANDI Archive.

AIND Ephys pipeline qualification conditions

The AIND Ephys pipeline runs on electrophysiology assets drawn from the DANDI Archive. To qualify, an asset must meet the following conditions:

1. The asset must be listed within a public Dandiset.
2. The asset must be an NWB file, either in HDF5 or Zarr format.
3. The NWB file must be openable and valid.
4. The NWB file must contain at least one `ElectricalSeries` data stream in the `acquisition` group with a `rate` greater than 10 kHz; lower-rate series (e.g. LFP) are ignored.

Each qualifying series must additionally:

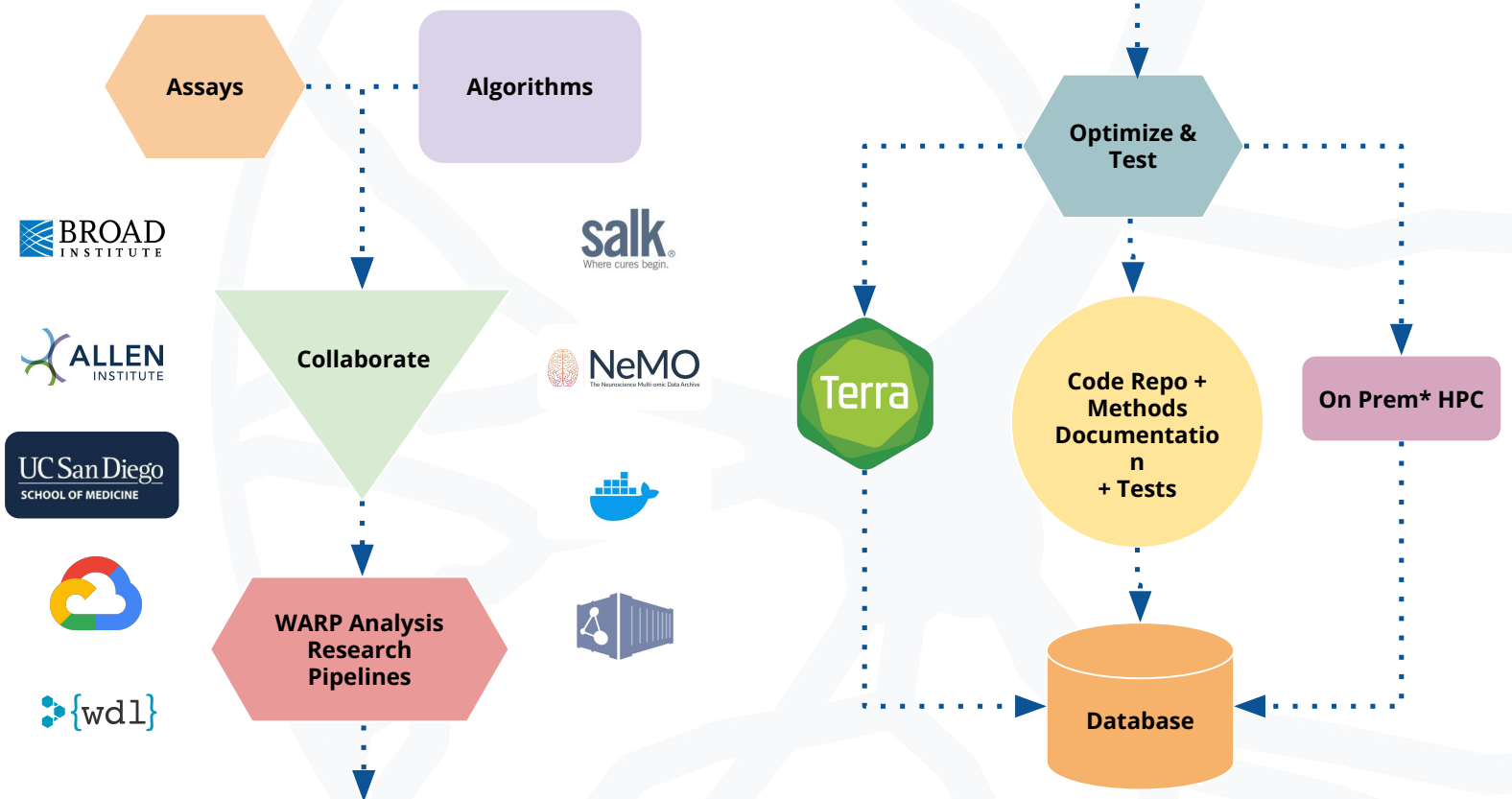
- have a total duration of more than 2 minutes.
- survive the pipeline's split-then-aggregate step: when a series spans more than one channel group, the pipeline splits series by channel group and recombines them with `spikeinterface.aggregate_channels`, which requires the relative channel locations to remain unique once combined.

See the full qualification conditions for the exact checks and common failure modes.

Explore & resources

RESOURCE	DESCRIPTION
Pipeline Results dashboard	Browse processing runs across Dandisets.
AIND Ephys pipeline	The electrophysiology processing pipeline itself.
DANDI Compute code	The compute codebase driving these runs.

Broad Pipeline Development: F.A.I.R. & Cloud-optimized Bioinformatic Pipelines

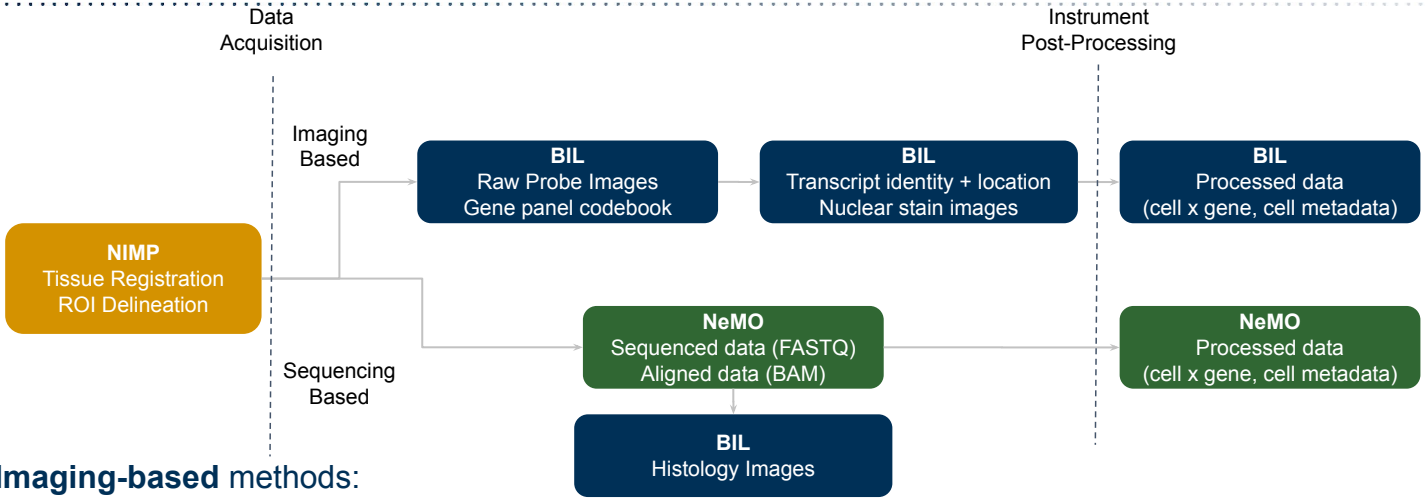


Pipelines to Process Different Assays and Modalities

Pipeline	WARP WDL Code	Assay Description	Publication
Single-cell ATAC (ATAC)	ATAC	Single-cell ATAC-seq data from nuclear isolates	Zhang et al., 2023
Multiome	Multiome	Multiplexed single-nucleus 3' gene expression and chromatin accessibility	
Optimus	Optimus	10x Genomics V2 and V3 3' single-cell and single-nucleus gene expression	
Single-nucleus Methylseq and 3C	snM3C	Multiplexed single-nucleus bisulfite sequencing and 3C data	Lee et al., 2019
Smart-seq (Full transcript RNA-seq, bulk)	Smart-seq	Single-nucleus RNA sequencing data generated with Smart-seq assays	

WARP - WDL Analysis Research Pipelines:
https://github.com/broadinstitute/warp

Spatial transcriptomics data workflow



Imaging-based methods:

1. Metadata entry at NIMP: Minimally enter donors and tissue sections and obtain NHASH IDs
2. Upload metadata to appropriate repository including NHASH IDs (or use backfill mechanism for previously uploaded data)
3. Upload data files to BIL

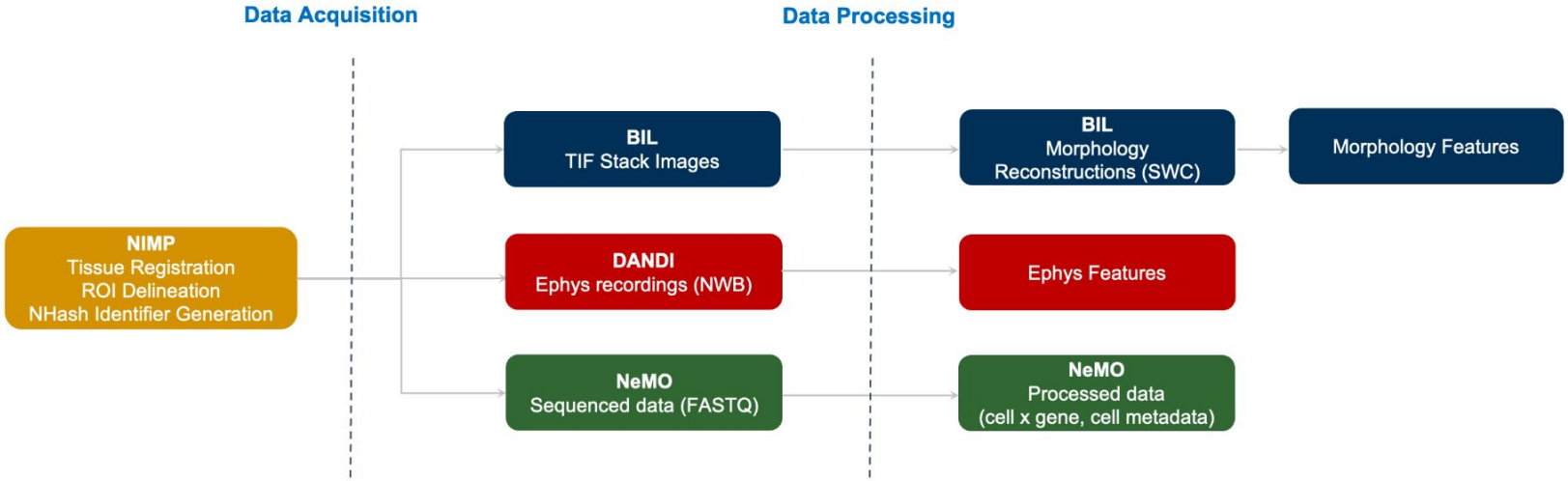
Sequencing based methods:

1. Generally expected to use sequencing workflow to have direct deposition of fastq at NeMO

Future:

1. Plan for improved integration of imaging and sequencing based data post-processing

Patch-seq data workflow



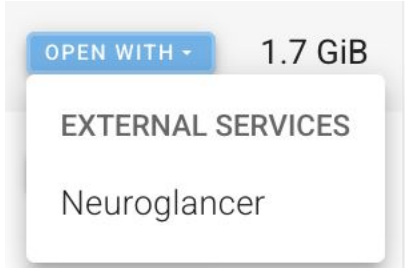
- Donor and specimen metadata to NIMP
  - Dissociated Cell Sample for cell-level specimen
- Electrophysiology metadata in NWB file
  - Schemas for experimental metadata in NWB files
- Transcriptomic Metadata to NeMO
  - NeMO submission manifest for non-pipeline tissue (contact NeMO)
- Morphology data to BIL
  - In BIL manifest for data uploads
  - For cell reconstructions, include module for SWC files

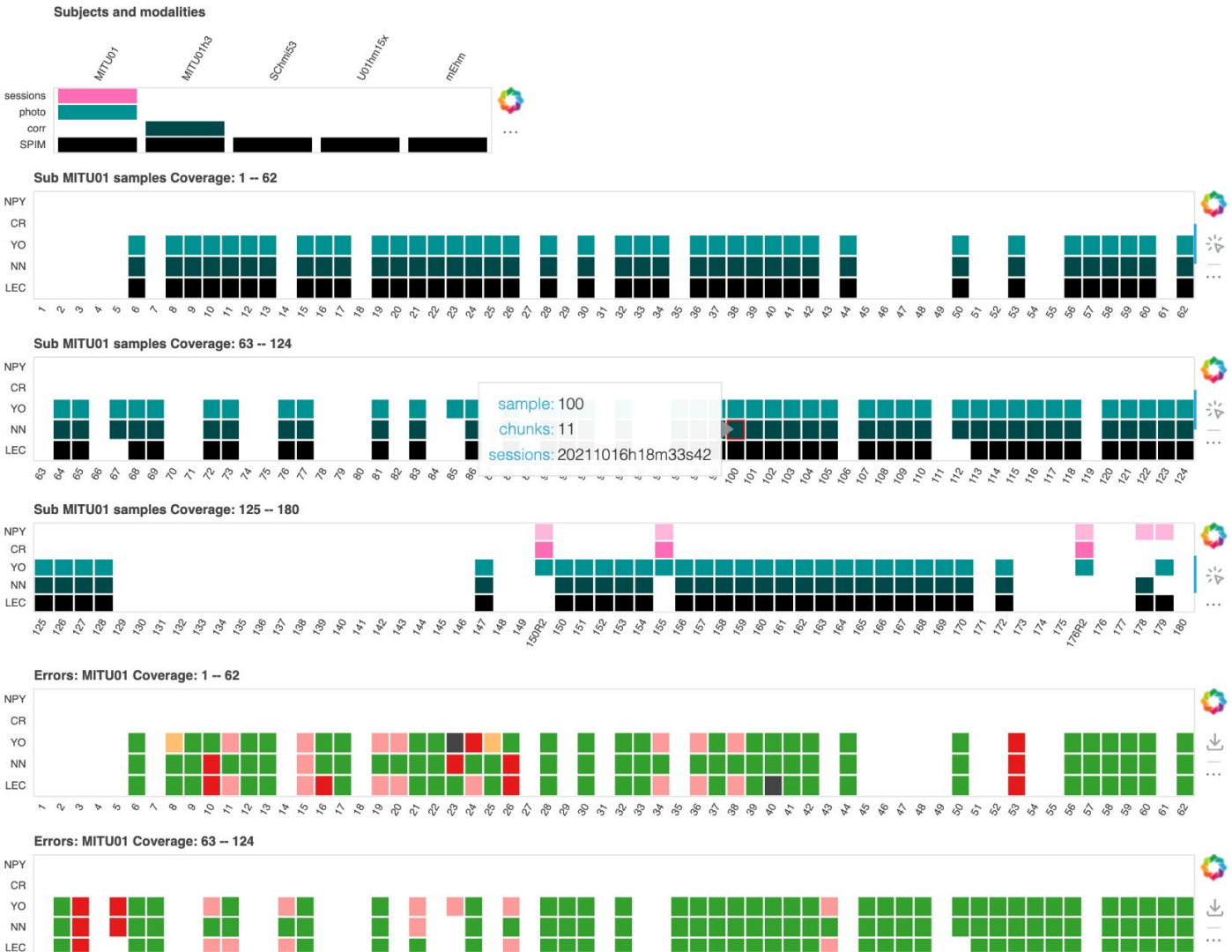
Proteomics data sharing

- How to share imaging data?
  - Convert to the OME-Zarr format (version 3)
  - Name according to the BIDS specification
  - Upload to DANDI Archive



- How can I visualize this data?
  - All open data on DANDI can be visualized with Neuroglancer. Navigate to the Zarr in the `File Browser` and select `Open With`.




Subjects and modalities
MITU01
MITU01h3
SChmi53
U01hm15x
mElm
sessions
photo
corr
SPIM
Sub MITU01 samples Coverage: 1 -- 62
NPY
CR
YO
NN
LEC
Sub MITU01 samples Coverage: 63 -- 124
sample: 100
chunks: 11
sessions: 20211016h18m33s42
Sub MITU01 samples Coverage: 125 -- 180
Errors: MITU01 Coverage: 1 -- 62
Errors: MITU01 Coverage: 63 -- 124

Knowledge resources

OPEN NEUROSCIENCE INFRASTRUCTURE

The open *neuroscience* knowledge graph.

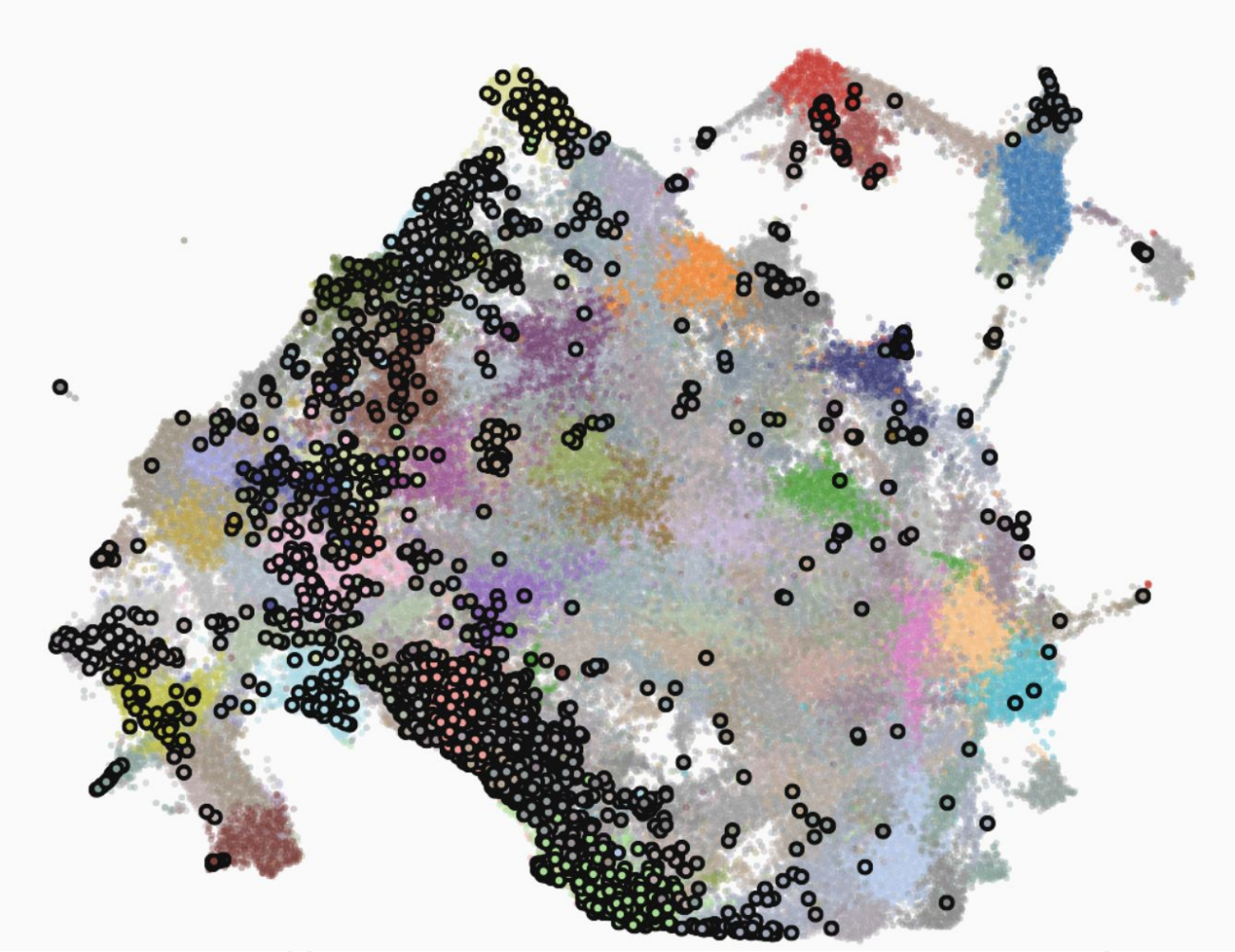
Trustworthy knowledge-graph infrastructure that integrates fragmented neuroscience literature, data, and evidence — enabling researchers to make reproducible discoveries and helping funders identify, evaluate, and invest in high-impact science.



8
open-source tools ↗

9
active use cases

Multi-agent
orchestration



https://abstractatlas.brainkb.org/

High-resolution Diffusion MRI

Reference

Gong et al., bioRxiv 2026

Data

DANDI Archive 001278 & 001372

Source Code

Repository on GitHub (Coming soon)

Acknowledgements

National Institutes of Health (NIH)
BRAIN CONNECTS program
UM1 NS132358

Explore the data

Human Ha1 data | Macaque M3 data

Layers

Axonal architecture

- [x] Isotropic component (l0)
- [x] Direction-encoded color (DEC) map
- [ ] Tract density

Tissue microstructure

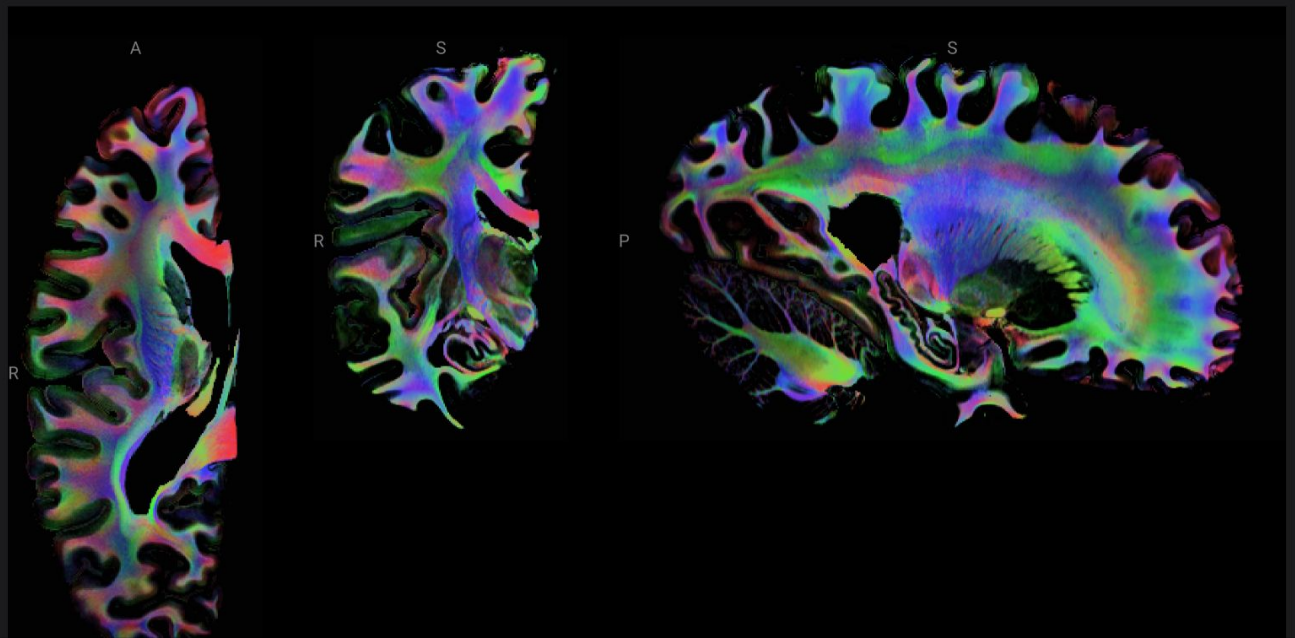
- [ ] Intra-soma signal fraction
- [ ] Intra-neurite signal fraction

Display options

- [ ] Crosshair

Mouse controls

Scroll slices	Scroll wheel
Move crosshair	Left click
Pan	Middle click
Zoom	Right click

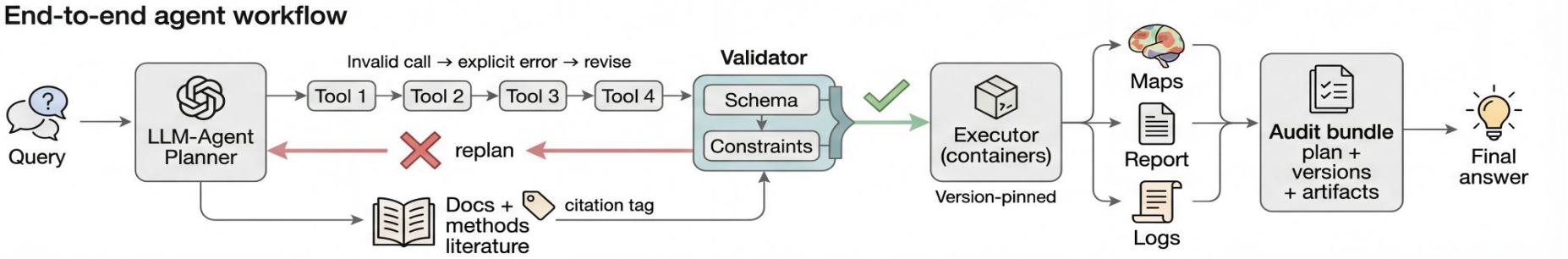


Brain Researcher MCP: An agent for reproducible neuroimaging analysis



Zijiao Chen

- A coding agent tool that generates a complete, executable analysis pipeline.
  - Every run produces a full provenance record: environment hash, input hash, exact commands, tool versions, output hash.
- Leverages a knowledge graph linking cognitive constructs, brain regions, analysis tools, valid parameters, and OpenNeuro datasets.
  - Formalizes the knowledge of domain experts



https://brain-researcher.com/

Brain Researcher

AI-assisted research infrastructure for neuroimaging

Turn neuroimaging questions into evidence-linked plans, runnable workflows, and reviewable scientific claims.

DATASETS	TOOL SPECS	MCP TOOLS	WORKFLOWS
1,600+	2,000+	87	60+

Tested across Claude, Codex, Gemini, GLM, DeepSeek, Kimi, and Qwen.

Claude Code | Codex | Cursor

1 Install Claude Code

npm install -g @anthropic-ai/claude-code

```
npm install -g @anthropic-ai/claude-co
```
Copy

2 Get your token

Sign in to mint your personal MCP token (a one-time secret).

3 Configure MCP

Set the token in your shell first. Then keep the MCP config as a separate file.

SHELL PROFILE ONLY

Add this to `~/.bashrc`, `~/.zshrc`, or `~/.profile`. Do not paste these export lines into `.mcp.json`.

```
# Brain Researcher MCP (add to ~/.bash
export BR_MCP_TOKEN="brk_<kid>.<secret>"
export BR_MCP_AUTH_HEADER="Bearer ${BR_MCP_TOKEN
```
Copy

Explore the resources

BRAIN Initiative PROJECTS
https://brain-bican.org
https://brain-connects.org
https://brain-bbqs.org

REPOSITORIES
https://bossdb.org
https://www.brainimagelibrary.org
https://dandiarchive.org
https://emberarchive.org
https://nemar.org
https://portal.nemoarchive.org
https://openneuro.org
https://knowledge.brain-map.org/data

STANDARDS
https://bids.neuroimaging.io
https://nwb.org
https://ngff.openmicroscopy.org

COMPUTATION/INTERACTION
https://neurosift.app
https://brain-researcher.com
https://dandi-compute.github.io/aind-page
https://gallery.lincbrain.org
https://biccn.github.io/Quarterly_Submission_Receipts/000108-dashboard.html

KNOWLEDGE RESOURCES
https://brain-map.org/bkp
https://abstractatlas.brainkb.org
https://brainkb.org